GLE
Facilities and Environmental
Consultants

February 1, 2012

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Mr. Jim Poje
US Fish and Wildlife Service
Division of Engineering, Room 330
Atlanta, GA, 30345

email: Jim_Poje@fws.gov

**RE: Comprehensive Asbestos Survey Report**
**Merritt Island National Wildlife Refuge**
**Titusville, FL**

GLE Project No.: 11463-00002

Dear Mr. Poje :

GLE Associates, Inc. (GLE) performed a comprehensive survey for asbestos-containing materials (ACM) on November 21-22, 2011 at the Merritt Island National Wildlife Refuge located in Titusville, Florida. The survey was performed by Mr. Johnny Ciucevich and Mr. Thomas Tripp with GLE. This report outlines the sampling and testing procedures, and presents the results along with our conclusions and recommendations.

GLE appreciates the opportunity to serve as your consultant on this project. If you should have any questions, or if we can be of further service, please do not hesitate to call.

Sincerely,
**GLE Associates, Inc.**

Thomas H. Tripp
Environmental Scientist

Robert B. Greene, PE, PG, CIH
Asbestos Consultant, EA 0000009

THT/MBC/RBG/lr

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# 1.0 EXECUTIVE SUMMARY

## 1.1 INTRODUCTION

The purpose of this comprehensive survey was to identify accessible asbestos-containing materials (ACMs) and their general locations within the Client-identified structures at the Merritt Island National Wildlife Refuge, located on SR 402 five (5) miles east of Titusville, Florida. The structures included in the survey are listed in the **Facility Description** below. **Note the Biolab Research Building had been demolished prior to the survey. There were no suspect materials at the Storage Shed at the Tranquility Helibase.** The survey was conducted pursuant to National Emission Standards for Hazardous Air Pollutants (NESHAP, 40 CFR 61) requirements associated with any future renovation or demolition plans. The survey was performed on November 21-22, 2011, by Mr. Johnny Ciucevich and Mr. Thomas Tripp, an Environmental Protection Agency/Asbestos Hazard Emergency Response Act (EPA/AHERA) accredited inspector. The scope of this survey did not include demolition of any building components, evaluation of architectural plans, or the quantification of materials for abatement purposes, or removal cost estimating.

## 1.2 FACILITY DESCRIPTION

A summary of the facilities investigated is outlined in the table below. A representative view of the facilities is shown in **Appendix E**.

**Biolab Well-Pump House**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | Unknown |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Wood Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Visitor Information Center**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1985 |
| Number of Floors: | 2 |
| **Structural** | |
| Foundation: | Crawl Space |
| Wall Support: | Wood Frame |
| Exterior Finish: | Pressboard Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | Air Handling Unit |
| Duct Type: | Metal, Flex Ductwork |
| Pipe Insulation: | Fiberglass |
| **Interior** | |
| Wall Substrate: | Drywall / Joint Compound |
| Wall Finishes: | Paint, Wallpaper |
| Floor Substrate: | Wood |
| Floor Finishes: | Carpet, Sheet Vinyl, Ceramic Tile, Wood |
| Ceiling System: | Drywall / Joint Compound |
| Ceiling Finishes: | Paint |

**Visitor Information Center Well-Pump House**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 2005 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Pressboard Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Office Trailer**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Pier & Beam |
| Wall Support: | Wood Frame |
| Exterior Finish: | Vinyl Siding |
| Roof Support: | Steel Joist |
| Roof System Type: | Composition Shingles, Rubber Membrane |
| **Mechanical/Plumbing** | |
| HVAC Type: | Bard Units |
| Duct Type: | Fiberglass Ductwork |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Drywall / Joint Compound |
| Wall Finishes: | Paint, Wallpaper |
| Floor Substrate: | Wood |
| Floor Finishes: | Carpet, Floor Tile |
| Ceiling System: | Suspended Ceiling System |
| Ceiling Finishes: | Suspended Ceiling Tiles |

**Light Engine Equipment Storage**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Wood Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | Bard Units |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Drywall / Joint Compound |
| Wall Finishes: | Paint |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished, Ceramic Tile |
| Ceiling System: | Drywall / Joint Compound |
| Ceiling Finishes: | Paint |

**Maintenance Shop**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 2 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Wood Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | Bard Units |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Drywall / Joint Compound |
| Wall Finishes: | Paint, Wood Paneling |
| Floor Substrate: | Concrete |
| Floor Finishes: | Paint, Sheet Vinyl |
| Ceiling System: | Drywall / Joint Compound, Suspended Ceiling System |
| Ceiling Finishes: | Paint, Suspended Ceiling Tiles |

**Fire Cache / Administrative Building**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1998 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Brick |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | Air Handling Unit |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Drywall / Joint Compound |
| Wall Finishes: | Paint |
| Floor Substrate: | Concrete |
| Floor Finishes: | Floor Tile |
| Ceiling System: | Drywall / Joint Compound |
| Ceiling Finishes: | Paint |

**Front Pole Shed**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | None Observed |
| Exterior Finish: | None Observed |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Back Pole Shed**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | None Observed |
| Wall Support: | None Observed |
| Exterior Finish: | None Observed |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | None Observed |
| Floor Finishes: | None Observed |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Service Building Fuel Island**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Concrete Masonry Units |
| Exterior Finish: | Concrete Masonry Units |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Concrete Masonry Units |
| Wall Finishes: | Paint |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished |
| Ceiling System: | Wood Paneling |
| Ceiling Finishes: | Paint |

**Warehouse**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1984 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Concrete Slab |
| Wall Support: | Wood Frame |
| Exterior Finish: | Wood Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | Wood Paneling |
| Wall Finishes: | Wood Paneling |
| Floor Substrate: | Concrete |
| Floor Finishes: | Concrete – Unfinished |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Engine Fill Station – Pump House**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1980 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | None Observed |
| Wall Support: | None Observed |
| Exterior Finish: | Plywood |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | None Observed |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | Gravel |
| Floor Finishes: | None Observed |
| Ceiling System: | None Observed |
| Ceiling Finishes: | None Observed |

**Pilot Briefing / Office Building**:

| | |
|---|---|
| Facility Type: | Commercial |
| Construction Date: | 1991 |
| Number of Floors: | 1 |
| **Structural** | |
| Foundation: | Pier & Beam |
| Wall Support: | Wood Frame |
| Exterior Finish: | Wood Siding |
| Roof Support: | Wood Truss |
| Roof System Type: | Composition Shingles |
| **Mechanical/Plumbing** | |
| HVAC Type: | Package A/C Unit (Window) |
| Duct Type: | None Observed |
| Pipe Insulation: | None Observed |
| **Interior** | |
| Wall Substrate: | None Observed |
| Wall Finishes: | None Observed |
| Floor Substrate: | Wood |
| Floor Finishes: | Sheet Vinyl |
| Ceiling System: | Wood Paneling |
| Ceiling Finishes: | Wood Paneling |

# 2.0 RESULTS

## 2.1 ASBESTOS SURVEY PROCEDURES

The comprehensive survey was performed by visually observing accessible areas of the building. EPA/AHERA accredited inspectors performed the visual observations (refer to Appendix B for personnel qualifications).

After the overall visual survey was completed, representative sampling areas were determined. The surveyors delineated homogeneous areas of suspect materials and samples of each material were obtained in general compliance with regulations as established by the Occupational Safety and Health Administration (OSHA) and NESHAP. The field surveyors determined sample locations based on previous experience. Both friable and non-friable materials were sampled. A friable material is one that can be crushed when dry by normal hand pressure. This survey did not include the demolition of building components to access suspect material.

After completion of the fieldwork, the samples were delivered to GLE's National Voluntary Laboratory Accreditation Program (NVLAP) accredited laboratory for analysis. The samples were analyzed by Polarized Light Microscopy (PLM) coupled with dispersion staining, in general accordance with EPA-600/R-93/116. Utilizing this procedure, the various asbestos minerals (chrysotile, amosite, crocidolite, actinolite, tremolite, and anthophyllite) can be determined. The percentages of asbestos minerals in the samples were visually determined by the microscopist. Please note that the EPA designates all materials containing greater than 1% asbestos as an "asbestos-containing material".

Regulated Asbestos-Containing Material (RACM) is defined as (a) Friable asbestos materials, (b) Category I non-friable ACM that has become friable, (c) Category I non-friable ACM that will be or has been subjected to sanding, grinding, cutting, or abrading, or (d) Category II non-friable ACM that has a high probability of becoming or has become crumbled, pulverized, or reduced to powder by the forces expected to act on the material in the course of demolition or renovation operations regulated by this subpart.

Category I and Category II non-friable ACM, as defined by the EPA:

- Category I non-friable ACM means asbestos containing packings, gaskets, resilient floor covering, and asphalt roofing products containing more than 1 percent asbestos and determined using the method specified in Appendix E, Subpart E, 40 CFR Part 763, Section 1, PLM.

- Category II non-friable ACM means any material, excluding Category I non-friable ACM, containing more than 1 percent asbestos as determined using the methods specified in Appendix E, Subpart E, 40 CFR Part 763 Section 1, PLM that, when dry, cannot be crumbled, pulverized, or reduced to powder by hand pressure.

## 2.2 IDENTIFIED SUSPECT ASBESTOS-CONTAINING MATERIALS

A total of 111 samples of suspect building materials were collected from the facilities during the survey, representing 37 different homogeneous areas. The results of the laboratory analyses are included in Appendix A, and approximate sample locations and the approximate extent to which ACM was observed to be present are indicated on the Asbestos Sample Location Plans and Asbestos Location Plans in Appendix C and D. Photographs of the various materials sampled are included in Appendix E.

A summary of the homogenous sampling areas of suspect ACM determined to be present is outlined in the following table:

**TABLE 2.2-1: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**BIOLAB WELL-PUMP HOUSE**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**TABLE 2.2-2: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**VISITOR INFORMATION CENTER**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| DW-01 | Drywall & Joint Compound | Throughout | NF | ND | 3 | NA | NA |
| MSV-01 | Beige Square Sheet Vinyl Flooring and Mastic | 2nd Floor Storage/Hallway | F | ND | 3 | NA | NA |
| **MSV-02** | **Brown Pebble Sheet Vinyl Flooring and Mastic** | **1st Floor Electrical Room & Mechanical Room** | **F** | **20% - Sheet Vinyl ND – Mastic** | **3** | **40 SF** | **RACM** |
| M-01 | Black Sink Undercoat | Break room | NF | ND | 3 | NA | NA |
| M-02 | Window Caulk | Exterior | NF | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**TABLE 2.2-3: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**VISITOR INFORMATION CENTER – WELL-PUMP HOUSE**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-4: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>OFFICE TRAILER<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| CT-01 | 2x4 Worm Hole Ceiling Tile | Throughout | F | ND | 3 | NA | NA |
| FT-01 | 12"x12" Tan Floor Tile and Mastic | Room by SE Entry Door | NF | ND | 3 | NA | NA |
| FT-02 | 12"x12" Cream w/ Blue Flake Floor Tile and Mastic | Restroom | NF | ND | 3 | NA | NA |
| DW-01 | Drywall & Joint Compound | Throughout | NF | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Porch Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-5: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>LIGHT ENGINE EQUIPMENT STORAGE<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| DW-01 | Drywall & Joint Compound | Offices 1&2 | NF | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**TABLE 2.2-6: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**MAINTENANCE SHOP**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| CT-01 | 2x2 Worm Hole Ceiling Tile | Break Room | F | ND | 3 | NA | NA |
| DW-01 | Drywall & Joint Compound | Offices / Break Room | NF | ND | 3 | NA | NA |
| MSV-01 | Tan & Blue Patterned Sheet Vinyl Flooring and Mastic | 2nd Floor Shower Room | F | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |
| **M-01** | **Window Glazing** | **Exterior Windows** | **NF** | **7%** | **3** | **1 SF** | **CAT II** |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**TABLE 2.2-7: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**FIRE CACHE / ADMINISTRATIVE BUILDING**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| DW-01 | Drywall & Joint Compound | Throughout | NF | ND | 3 | NA | NA |
| M-01 | White Sink Undercoat | Kitchen | NF | ND | 3 | NA | NA |
| FT-01 | 12"x12" Tan w/ Brown Flake Floor Tile and Mastic | Throughout | NF | ND | 3 | NA | NA |
| FT-02 | 12"x12" Brown Small Square Floor Tile and Mastic | Bathrooms 1&2 | NF | ND | 3 | NA | NA |
| FT-03 | 12"x12" White Floor Tile and Mastic | By Back Door | NF | ND | 3 | NA | NA |
| M-02 | Window Caulk | Exterior Window | NF | ND | 3 | NA | NA |
| M-03 | Hurricane Shutter Caulk | Exterior Shutters | NF | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-8: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>FRONT POLE SHED<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-9: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>BACK POLE SHED<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-10: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>SERVICE BUILDING FUEL ISLAND BUILDING<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-11: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>WAREHOUSE<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |
| MAS-01 | Pipe Mastic | Throughout | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**TABLE 2.2-12: SUMMARY OF HOMOGENEOUS SAMPLING AREAS**
**ENGINE FILL STATION – PUMP HOUSE**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| HA # | HOMOGENEOUS MATERIAL DESCRIPTION | HOMOGENEOUS MATERIAL LOCATION | FRIABILITY (F /NF) | % ASBESTOS* | # OF SAMPLES COLLECTED | APPROXIMATE QUANTITY | ACM CATEGORY |
|---|---|---|---|---|---|---|---|
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT** Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

| TABLE 2.2-13: SUMMARY OF HOMOGENEOUS SAMPLING AREAS<br>PILOT BRIEFING / OFFICE BUILDING<br>MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL | | | | | | | |
|---|---|---|---|---|---|---|---|
| **HA #** | **HOMOGENEOUS MATERIAL DESCRIPTION** | **HOMOGENEOUS MATERIAL LOCATION** | **FRIABILITY (F /NF)** | **% ASBESTOS*** | **# OF SAMPLES COLLECTED** | **APPROXIMATE QUANTITY** | **ACM CATEGORY** |
| M-01 | Window Caulk | Exterior Windows | NF | ND | 3 | NA | NA |
| MSV-01 | Blue Speck Sheet Vinyl Flooring and Mastic | Throughout | F | ND | 3 | NA | NA |
| RS-01 | Roof Shingles | Roof | NF | ND | 3 | NA | NA |

| | | | | |
|---|---|---|---|---|
| **ASBESTOS CONTENT**<br>Expressed as percent | * = The facility owner has the option of point-counting by polarized light microscopy (PLM) those RACM whose asbestos content is less than 10% in order to more accurately determine the asbestos content therein. | | | |
| | PC = Results based on Point-Count analysis | | | |
| **FRIABILITY** | F = Friable Material | NF = Non-Friable Material | | |
| **CATEGORY OF MATERIAL** | RACM = Regulated ACM | CAT I = Category I non-friable ACM | CAT II = Category II non-friable ACM | |
| **ABBREVIATIONS:** | NA = Not Applicable | ND = None Detected | NIS = Not in Scope | |
| | HA = Homogeneous Area | SF = Square Feet | LF = Linear Feet | CF = Cubic Feet |

**Due to the damage that could be caused by destructive sampling techniques and restricted accessibility, the following materials were assumed to contain asbestos minerals and were not sampled:**

1. **Rubber Membrane Roofing; (Office Trailer). Not sampled due to the Standards of Operation.**

# 3.0 CONCLUSIONS AND RECOMMENDATIONS

## 3.1 GENERAL

The EPA, OSHA and the have promulgated regulations dealing with asbestos. For commercial building owners, the EPA NESHAP (40 CFR 61) regulations require removal of RACM prior to conducting activities which might disturb the material. They also deal with notification, handling and disposal of asbestos.

No homogenous areas of suspect RACM was determined to contain less than 10% asbestos by PLM analysis. According to the NESHAP, when the asbestos content of a bulk sample of suspect RACM is determined to be less than 10% by PLM visual estimation, you may:

1. Assume the amount to be greater than 1% and treat the material as asbestos-containing; or
2. Conduct confirmatory verification by point-counting. Note, the results obtained by point-counting are considered the definitive analytical result.

For facilities not scheduled for demolition or complete removal of all ACM, the EPA recommends that an Operations and Maintenance (O&M) Program be developed for any facilities with ACM, and this Program should address all ACM (known and/or assumed) present. The O&M Program establishes notification and training requirements along with special procedures for working around the asbestos. The O&M Program would remain in effect until all asbestos is removed.

Category I and Category II non-friable materials, as defined by the EPA, may remain within a facility during demolition with no potential cessation of work provided they remain non-friable and the appropriate engineering controls (i.e., wet methods) are utilized, with the resulting waste disposed of as asbestos-containing waste. However, there is no guarantee that these materials will remain non-friable. If the materials become friable, then they are classified as RACM.

RACM, as defined by the EPA, must be removed prior to renovation or demolition activities that may disturb the materials.

The OSHA regulations deal with employee exposure to airborne asbestos fibers. The regulations restrict employee exposure, and require special monitoring, training and handling procedures when dealing with asbestos. Additionally, OSHA has regulations that may supersede the EPA regulations. In order to protect the worker, OSHA has established a permissible exposure limit (PEL), which limits employee exposure to airborne fiber concentrations. OSHA requires

objective evidence that the PEL will not be exceeded, as justification that personal air monitoring and engineering controls will not be required. OSHA has also established rules requiring the containerization and labeling of asbestos waste.

The State regulations require that anyone involved in asbestos consulting activities be a licensed asbestos consultant and that anyone involved in asbestos abatement, with the exception of roofing materials, be a licensed asbestos abatement contractor.

## 3.2 SPECIFIC

### Window Glazing (Maintenance Shop)

This material is defined by the EPA as a Category II non-friable material. This material does not appear to present a significant issue as observed at the time of the survey. We recommend that the identified ACM be maintained as part of an O&M Program and periodically monitored for any changes in condition. Additionally, we recommend that a licensed asbestos abatement contractor properly remove the ACM prior to conducting renovation activities that might disturb the ACM.

### Brown Pebble Sheet Vinyl Flooring (Visitors Information Center)

This material is defined by the EPA as RACM. This material does not appear to present a significant issue as observed at the time of the survey. We recommend that the identified RACM be maintained as part of an O&M Program and periodically monitored for any changes in condition. Additionally, we recommend that a licensed asbestos abatement contractor properly remove the identified RACM prior to conducting renovation activities that might disturb the ACM.

### Rubber Membrane Roof (Office Trailer)

The assumed asbestos-containing Membrane Roof observed associated with the Office Trailer appeared to be in generally good condition and does not present a significant issue as observed at the time of the survey. We recommend that this material be maintained as part of an O&M Program and periodically monitored for any changes in condition. Additionally, should planned renovation and/or demolition activities involve the disturbance of this material, we recommend that it be sampled and analyzed for asbestos content, and if determined to be ACM, properly removed by a licensed asbestos abatement contractor prior to conducting such activities.

## 3.3 ASBESTOS REMOVAL OPINION OF COST

The following opinion of cost can be used for determining approximate costs associated with the removal of ACM identified during the survey. Please note that the quantities outlined below reflect a combination of field verification of material locations, provided as-built documentation or previous survey experience of similar structures. Actual field verification was limited during the survey as a result of accessibility. The opinion of cost reflect typical industry standard

removal rates, but can vary dependent upon the number of mobilizations required of the contractor, the timing of the project and accessibility restrictions to the material to be abated during the performance of these activities.

**TABLE 3.3-1: OPINION OF COST**
**VISITORS INFORMATION CENTER**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| Homogeneous Area | | Approximate Quantity | Unit Cost | Opinion of Cost |
|---|---|---|---|---|
| Number | Description | | | |
| MSV-02 | Brown Pebble Sheet Vinyl Flooring | 40 SF | $20/SF | $800 |
| | | | TOTAL | $800 |

SF: square feet LF: linear feet CF: cubic feet

**TABLE 3.3-2: OPINION OF COST**
**MAINTENANCE SHOP**
**MERRITT ISLAND NATIONAL WILDLIFE REFUGE – TITUSVILLE, FL**

| Homogeneous Area | | Approximate Quantity | Unit Cost | Opinion of Cost |
|---|---|---|---|---|
| Number | Description | | | |
| M-01 | Window Glazing | 1 SF (1 Window) | $500/window | $500 |
| | | | TOTAL | $500 |

SF: square feet LF: linear feet CF: cubic feet

The opinion of cost provided assumes that much of the work will be performed in conjunction with any renovation activities, and that much of the demolition work required to gain access to the ACM is not part of the abatement contractor's work. **Also, no opinion of cost was provided for the removal of roofing materials, as a properly licensed and qualified roofing contractor and not an asbestos abatement contractor typically performs this activity.** In addition to the opinion of cost outlined above, typical consulting fees for project design, contract administration, project monitoring and laboratory fees add 20-40% to removal costs for projects of similar size and expected duration.

# 4.0 LIMITATIONS AND CONDITIONS

As a result of previous renovations, there may be hidden materials, such as floor tile, sheet vinyl flooring, etc. These materials may be found in various areas hidden under existing flooring materials. Any materials found during construction activities, either not addressed in this survey report, or similar to the ACM identified in this survey report should be assumed to be ACM until sampling and analysis documents otherwise.

The Florida Department of Environmental Protection (FDEP) has recently issued an interpretation regarding the testing of concrete flooring, walls and roofing materials, which states that "if concrete will be recycled or reused, the concrete must be sampled and analyzed for the presence of asbestos prior to the commencement of activities that may release asbestos fibers into the environment", and that "all of the different layers or types of concrete in a sample must be analyzed, individually, using the method specified in Appendix E, subpart E, 40 CFR Part 763, Section 1, Polarized Light Microscopy, with point-counting", as applicable. Under the presumption that the Client will not be reusing/recycling the concrete, this additional sampling and analysis of concrete is not included with our scope of work. However, if requested by the Client, GLE will perform this work as an additional service.

Because of the hidden nature of many building components (i.e. within mechanical chases), it may be impossible to determine if all of the suspect building materials have been located and subsequently tested. Destructive testing in some instances is not a viable option. We cannot, therefore, guarantee that all potential ACM has been located. For the same reasons, estimates of quantities and/or conditions are subject to readily apparent situations, and our findings reflect this condition. We do warrant, however, that the investigations and methodology reflect our best efforts based upon the prevailing standard of care in the environmental industry.

The information contained in this report was prepared based upon specific parameters and regulations in force at the time of this report. The information herein is only for the specific use of the client and GLE. GLE accepts no responsibility for the use, interpretation, or reliance by other parties on the information contained herein, unless prior written authorization has been obtained from GLE.

# APPENDIX A
# Analytical Results and Chains of Custody

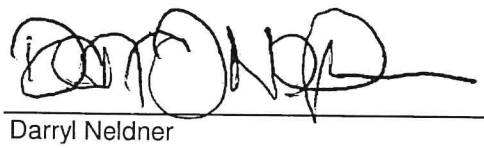
# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Bio Lab Well Pump House

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:



Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14653

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Bio Lab Well Pump House

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
| | | | |
| | | | |
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| | | | |
| | | | |

| | |
|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____/_____ **SAMPLE ANALYSIS DEADLINE**
**date / time**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: FedEx | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Asb Survey - Visitor Information Center

11463-00002

| Sample | Sample Type | | Fiber Type |
|---|---|---|---|
| DW-01A | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01B | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01C-QC | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| M-01A | Black Sink Undercoat | 100% | Bitumen, Quartz, Calcite, Mica |
| M-01B | Black Sink Undercoat | 100% | Bitumen, Quartz, Calcite, Mica |
| M-01C | Black Sink Undercoat | 100% | Bitumen, Quartz, Calcite, Mica |
| M-02A | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-02B | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-02C | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| MSV-01A | Beige Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01B | Beige Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01C | Beige Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-02A-QC | Brown Sheet Vinyl | **20%**<br>80% | **Chrysotile Asbestos**<br>Polymer |
| | Tan Mastic | 100% | Polymer |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14647

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Asb Survey - Visitor Information Center

11463-00002

| Sample | Sample Type | | Fiber Type |
|---|---|---|---|
| MSV-02B | Brown Sheet Vinyl | | Positive Stop/Sample not analyzed |
| | Tan Mastic | 100% | Polymer |
| MSV-02C | Brown Sheet Vinyl | | Positive Stop/Sample not analyzed |
| | Tan Mastic | 100% | Polymer |
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14647

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Visitor Information Center

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| DW-01-A-C | Drywall / Joint Compound | | |
| MSV-01-A-C | Beige Square Sheet Vinyl Flooring | | |
| MSV-02-A-C | Brown Pebble Sheet Vinyl Flooring | | |
| M-01-A-C | Black Sink Undercoat | | |
| M-02-A-C | Window Caulk | | |
| RS-01-A-C | Roof Shingles | | |
| | | | |
| | | | |
| | | | |
| | | | |
| | | | |

| | |
|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 18 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____/_____ **SAMPLE ANALYSIS DEADLINE**
date / time

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: JC M. Tripp | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: FED EX | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Visitor Ctr Well Pump House

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14658

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Visitor Center Well Pump House

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: AL H. [illegible] | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: FedEx | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Office Trailer

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| CT-01A | 2x4 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| CT-01B | 2x4 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| CT-01C-QC | 2x4 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| DW-01A | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01B | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01C | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| FT-01A | 12x12 Tan Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-01B | 12x12 Tan Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-01C | 12x12 Tan Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02A | 12x12 Cream/Blue Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02B | 12x12 Cream/Blue Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02C | 12x12 Cream/Blue Floor Tile & Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14648

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Office Trailer

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14648

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

1462

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Office Trailer

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
| CT-01-A-C | 2x4 Worm Hole Ceiling tile | | |
| FT-01-A-C | 12x12 Tan Floor Tile & Mastic | | |
| FT-02-A-C | 12x12 Cream & Blue Flake Floor Tile & Mastic | | |
| DW-01-A-C | Drywall & Joint Compound | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 15 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: TC H. TP | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: F-d EX | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Light Engine - Equip Storage

11463-00002

| Sample | Sample Type | | Fiber Type |
|---|---|---|---|
| DW-01A | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01B | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01C-QC | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| RS-01A | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14651

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey -- Light Engine / Equipment Storage

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| DW-01-A-C | Drywall & Joint Compound | | |
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 6 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: TC H Tapp | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fedex | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Maintenance Shop

11463-00002

| Sample | Sample Type | | Fiber Type |
|---|---|---|---|
| CT-01A | 2x2 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| CT-01B | 2x2 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| CT-01C-QC | 2x2 Worm Hole Ceiling Tile | 70%<br>30% | Mineral Wool<br>Perlite, Quartz, Calcite |
| DW-01A | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01B | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01C | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| M-01A | Window Glazing | **7%**<br>93% | **Chrysotile Asbestos**<br>Polymer, Quartz, Calcite, Clay, Mica |
| M-01B | Window Glazing | | Positive Stop/Sample not analyzed |
| M-01C | Window Glazing | | Positive Stop/Sample not analyzed |
| MSV-01A | Tan/Blue Patterned Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01B | Tan/Blue Patterned Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01C | Tan/Blue Patterned Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14649

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Maintenance Shop

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Mineral Wool |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14649

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Maintenance Shop

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| DW-01-A-C | Drywall & Joint Compound | | |
| M-01-A-C | Window Glazing | | |
| CT-01-A-C | 2x2 Ceiling Tile Worm Hole | | |
| MSV-01-A-C | Tan & Blue Patterned Sheet Vinyl Flooring | | |
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | **15** |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____/_____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: Tim M. Tripp | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: FedEx | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

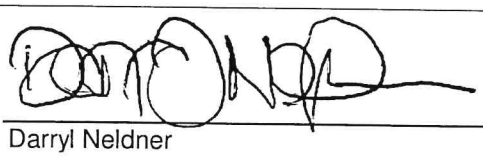
# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Fire Cache Admin Bulding

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| DW-01A | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01B | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| DW-01C-QC | Drywall & Joint Compound | 100% | Gypsum, Quartz, Calcite, Clay |
| FT-01A | 12x12 Tan/Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-01B | 12x12 Tan/Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-01C | 12x12 Tan/Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02A | 12x12 Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02B | 12x12 Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-02C | 12x12 Brown Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-03A | 12x12 White Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-03B | 12x12 White Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| FT-03C | 12x12 White Floor Tile & Tan Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-01A-QC | White Sink Undercoat | 100% | Polymer, Quartz, Calcite, Clay, Mica |

Analyst / Approved Signatory:



Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14646

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

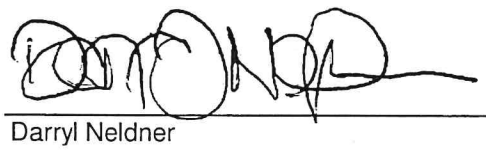
# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Fire Cache Admin Bulding

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| M-01B | White Sink Undercoat | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-01C | White Sink Undercoat | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-02A | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-02B | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-02C | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-03A | Hurricane Shutter Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-03B | Hurricane Shutter Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-03C | Hurricane Shutter Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| RS-01A | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01B-QC | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:



Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14646

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Fire Cache Administrative Building

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| DW-01-A-C | Drywall & Joint Compound | | |
| M-01-A-C | White Sink Undercoat | | |
| FT-01-A-C | 12x12 Tan w/ Brown Flake Floor Tile | | |
| FT-02-A-C | 12x12 Brown Small Square Floor Tile | | |
| FT-02-A-C | 12x12 White Floor Tile | | |
| M-02-A-C | Window Caulk | | |
| M-03-A-C | Hurricane Shutter Caulk | | |
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 24 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____ **SAMPLE ANALYSIS DEADLINE**
**date / time**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: H. Ty | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fed EX | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Front Pole Shed

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14655

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

LAB 14655

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Front Pole Shed

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: JC N. TB | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: FED EX | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

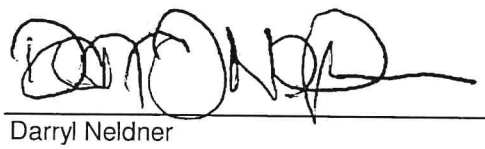
# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Back Pole Shed

11463-00002

| Sample | Sample Type | Fiber Type |
|---|---|---|
| RS-01A | Roof Shingles | 10% Glass Fibers<br>90% Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% Glass Fibers<br>90% Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% Glass Fibers<br>90% Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:



Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14656

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

## CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Back Pole Shed

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

### SAMPLE INFORMATION

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

### SAMPLE INSTRUCTIONS

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

### REPORT RESULTS TO THE ADDRESS ABOVE

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: TL M.TP | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: fedex | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Fuel Island Building

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14654

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Fuel Island Building

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____/_____ **SAMPLE ANALYSIS DEADLINE**
date / time

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: TL H. TJB | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fedex | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Warehouse

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| MAS-01A | Pipe Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| MAS-01B | Pipe Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| MAS-01C-QC | Pipe Mastic | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| RS-01A | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14652

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Warehouse

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
| MAS-01-A-C | Pipe Mastic | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 6 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____/_____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fedex | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Engine Fill Station

11463-00002

| Sample | Sample Type | Fiber Type | |
|---|---|---|---|
| RS-01A | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |
| RS-01C-QC | Roof Shingles | 10% | Glass Fibers |
| | | 90% | Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14657

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Engine Fill Station

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 3 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fed ex | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# SUMMARY OF BULK SAMPLE ANALYSIS

## U.S. Fish & Wildlife Services; Pilot Briefing/Office Building

11463-00002

| Sample | Sample Type | | Fiber Type |
|---|---|---|---|
| M-01A | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-01B | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| M-01C-QC | Window Caulk | 100% | Polymer, Quartz, Calcite, Clay, Mica |
| MSV-01A | Blue Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01B | Blue Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| MSV-01C | Blue Sheet Vinyl & Mastic | 20%<br>80% | Cellulose/paper<br>Polymer |
| RS-01A | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01B | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |
| RS-01C | Roof Shingles | 10%<br>90% | Glass Fibers<br>Bitumen, Quartz, Calcite, Mica |

Analyst / Approved Signatory:

Darryl Neldner

* Polarized Light Microscopy coupled with dispersion is the technique used for identification in accordance with EPA 600/M4-82-020, EPA 600/R-93/116, and NIOSH Method 9002.

** The percentage of each component is visually estimated. The result of this analysis relate only to the material tested. The report shall not be used to claim product endorsement by NVLAP or any agency of the U.S. Government. (>1% greater than one percent, <1% less than one percent) QC - Sample reanalyzed for QA/QC.

*** This report shall not be reproduced except in full, without the written approval of the laboratory. GLE Report # 14650

Analysis performed by GLE Associates, Inc. NVLAP #102003-0, CO AL-17485, TX 30-0337

Feedback regarding laboratory performance should be addressed to lab@gleassociates.com.

**CHAIN OF CUSTODY/SAMPLE TRANSMITTAL FORM**

GLE Associates, Inc.
4300 W. Cypress Street, Suite 400
Tampa, FL 33607
Tel. (813) 241-8350 FAX (813) 241-8737

**CLIENT:** U.S. Fish & Wildlife Services

14650

**PROJECT #:** 11463-00002

**PROJECT:** Merritt Island National Wildlife Refuge – Asbestos Survey – Pilot Briefing / Office Building

**LABORATORY SENT TO:** GLE – Tampa

**DATE:** 11-23-11

**SAMPLE INFORMATION**

| SAMPLE # | DESCRIPTION/ LOCATION | SAMPLE # | DESCRIPTION/ LOCATION |
|---|---|---|---|
| RS-01-A-C | Roof Shingles | | |
| M-01-A-C | Window Caulk | | |
| MSV-01-A-C | Blue Speck Sheet Vinyl Flooring | | |
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|---|---|
| **IMPORTANT TOTAL NUMBER OF SAMPLES SUBMITTED:** | 9 |
| **IMPORTANT POSITIVE STOP ANALYSIS:** | YES |
| **IMPORTANT CODE TYPE (PLM; PLM1; PLM 2; ETC.):** | PLM9 / 5day + |
| **IMPORTANT E-MAIL RESULTS TO:** | Pzac, jciucevich, ttripp |

**SAMPLE INSTRUCTIONS**

TO BE ANALYZED FOR ASBESTOS CONTENT BY POLARIZED LIGHT MICROSCOPY WITH DISPERSION STAINING

TURNAROUND TIME DEADLINE

→ → RETURN SAMPLES TO GLE ASSOCIATES USE TRANSMITTAL

_____ / _____
date / time

**SAMPLE ANALYSIS DEADLINE**

**REPORT RESULTS TO THE ADDRESS ABOVE**

| CHAIN OF CUSTODY: GLE ASSOCIATES, INC. | CHAIN OF CUSTODY: LABORATORY |
|---|---|
| PACKAGED BY: JL H. TB | SAMPLES RECEIVED BY: |
| DATE PACKAGED: 11-23-11 | DATE: |
| METHOD OF TRANSMITTAL: Fed EX | TIME: |
| TRANSMITTED BY: | CONDITION OF PACKAGED SAMPLES: |

| CHAIN OF CUSTODY: RETURNED TO GLE ASSOCIATES, INC. | |
|---|---|
| RECEIVED BY: | DATE: |
| INVENTORIED BY: | DATE: |
| REPACKAGED AND SEALED BY: | DATE: |

PAGE: OF

# APPENDIX B
# Personnel and Laboratory Certifications

**STATE OF FLORIDA**

DEPARTMENT OF BUSINESS AND PROFESSIONAL REGULATION

ASBESTOS LICENSING UNIT
1940 NORTH MONROE STREET
TALLAHASSEE FL 32399-0783

(850) 487-1395

GREENE, ROBERT BLAIR
GLE ASSOCIATES INC
4300 W CYPRESS STREET SUITE 400
TAMPA FL 33607

Congratulations! With this license you become one of the nearly one million Floridians licensed by the Department of Business and Professional Regulation. Our professionals and businesses range from architects to yacht brokers, from boxers to barbeque restaurants, and they keep Florida's economy strong.

Every day we work to improve the way we do business in order to serve you better. For information about our services, please log onto **www.myfloridalicense.com**. There you can find more information about our divisions and the regulations that impact you, subscribe to department newsletters and learn more about the Department's initiatives.

Our mission at the Department is: License Efficiently, Regulate Fairly. We constantly strive to serve you better so that you can serve your customers. Thank you for doing business in Florida, and congratulations on your new license!

STATE OF FLORIDA AC# 5348917
DEPARTMENT OF BUSINESS AND PROFESSIONAL REGULATION

EA0000009 11/19/10 108130228

ASBESTOS CONSULTANT - ENGINEER
GREENE, ROBERT BLAIR
GLE ASSOCIATES INC

IS LICENSED under the provisions of Ch.469 FS.
Expiration date: NOV 30, 2012 L10111901357

DETACH HERE

AC# 5348917

**STATE OF FLORIDA**

DEPARTMENT OF BUSINESS AND PROFESSIONAL REGULATION
ASBESTOS LICENSING UNIT

SEQ# L10111901357

| DATE | BATCH NUMBER | LICENSE NBR |
|---|---|---|
| 11/19/2010 | 108130228 | EA0000009 |

The ASBESTOS CONSULTANT - ENGINEER
Named below IS LICENSED
Under the provisions of Chapter 469 FS.
Expiration date: NOV 30, 2012

GREENE, ROBERT BLAIR
GLE ASSOCIATES INC
3109 W DR MARTIN LUTHER KING JR BLV
SUITE 550
TAMPA FL 33607

CHARLIE CRIST
GOVERNOR

CHARLIE LIEM
SECRETARY

DISPLAY AS REQUIRED BY LAW

# M·E·T·A

*Mayhew Environmental Training Associates*

INCORPORATED

Certificate # 7ME02081102AIR0011

*This is to certify that*

# John Ciucevich

*has on 2/8/11, in Jacksonville, FL*
*completed the requirements for asbestos accreditation under Section 206 of TSCA Title II, 15 U.S.C. 2646*

## AHERA Asbestos Building Inspector Refresher Course

*as approved by the State of Florida and the U.S.E.P.A. under 40 C.F.R. 763 (AHERA)*
*on 2/8/11 - 2/8/11 and passed the associated examination on 2/8/11*
*with a score of 70% or better*
*CM = 0.5*

KANSAS MAYHEW EST. 1985 INC. 1987 ENVIRONMENTAL TRAINING ASSOCIATES SAFETY IN GOD ALONE

Provider #: FL49-0001221
Course #: FL49-0004718

Soc. Sec #: XXX-XX-9848
Accreditation Expires: 2/8/12

*Instructor*
*Bill Young*

*President*
*Thomas Bradford Mayhew*

M E T A - P.O. Box 786 - Lawrence KS 66044 - 800-444-6382

# M·E·T·A

*Mayhew Environmental Training Associates*

INCORPORATED

Certificate # 7ME03141106AI00006

*This is to certify that*

# Thomas Tripp

*has on 3/16/11, in Jacksonville, FL*
*completed the requirements for asbestos accreditation under Section 206 of TSCA Title II, 15 U.S.C. 2646*

## AHERA Asbestos Inspector Training

*as approved by the State of Florida and the U.S.E.P.A. under 40 C.F.R. 763 (AHERA)*
*on 3/14/11 - 3/16/11 and passed the associated examination on 3/16/11*
*with a score of 70% or better*
*CM = 3.0*

KANSAS MAYHEW EST. 1985 INC. 1987 ENVIRONMENTAL TRAINING ASSOCIATES SAFETY IN GOD ALONE

Provider #: FL49-0001221
Course #: FL49-0002839

Soc. Sec.#: XXX-XX-1299
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United States Department of Commerce
National Institute of Standards and Technology

NVLAP®

# Certificate of Accreditation to ISO/IEC 17025:2005

**NVLAP LAB CODE: 102003-0**

## GLE Associates, Inc.

Tampa, FL

*is accredited by the National Voluntary Laboratory Accreditation Program for specific services, listed on the Scope of Accreditation, for:*

## BULK ASBESTOS FIBER ANALYSIS

*This laboratory is accredited in accordance with the recognized International Standard ISO/IEC 17025:2005. This accreditation demonstrates technical competence for a defined scope and the operation of a laboratory quality management system (refer to joint ISO-ILAC-IAF Communique dated January 2009).*

2011-04-01 through 2012-03-31

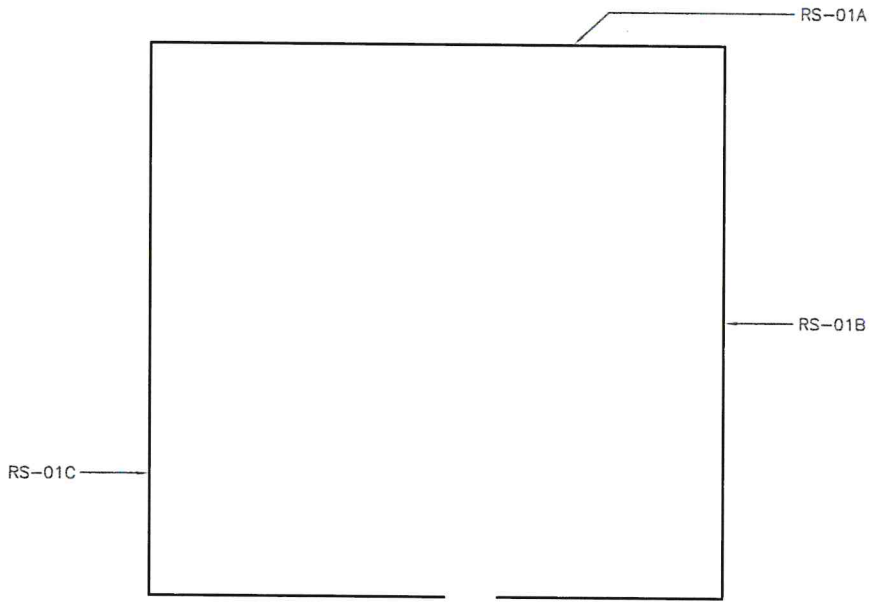
*Effective dates*

DEPARTMENT OF COMMERCE ★ UNITED STATES OF AMERICA ★

Sally S. Bruce

*For the National Institute of Standards and Technology*

# APPENDIX C
# Asbestos Sample Location Plans




ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
BIOLAB WELL PUMP HOUSE
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

Prepared for:
U.S. FISH & WILDLIFE SERVICES
P.O. BOX 2683
TITUSVILLE, FL

Prepared By: GLE ASSOCIATES, INC.
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TAMPA, FLORIDA 33607
(813) 241-8350 (813) 241-8737

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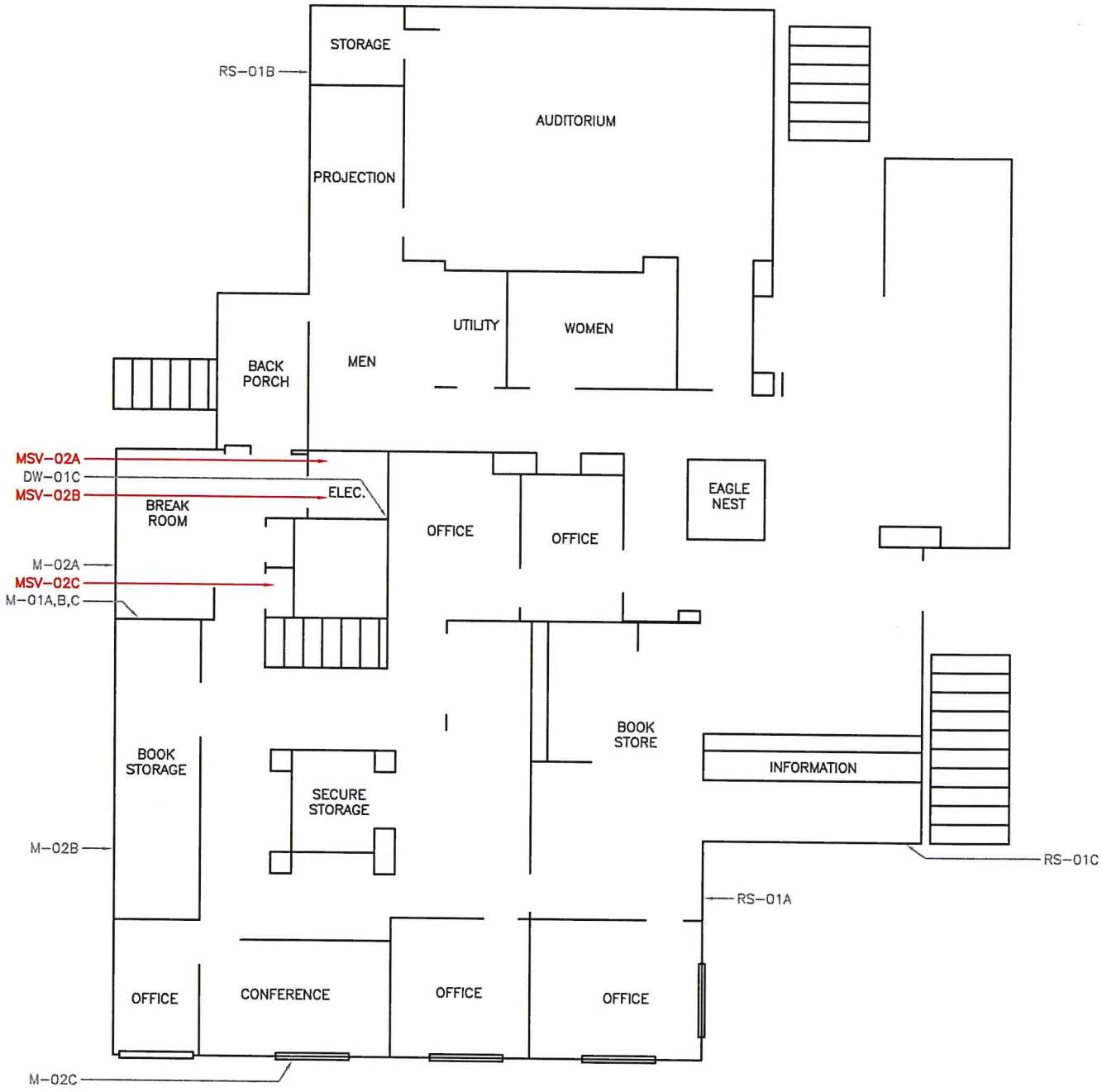
| DATE: 12/01/11 | OF 15 SHEET(S) |




ASBESTOS CONTAINING MATERIALS LEGEND

MSV-02: BROWN PEBBLE SHEET VINYL FLOORING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
VISITOR INFORMATION CENTER – 1ST FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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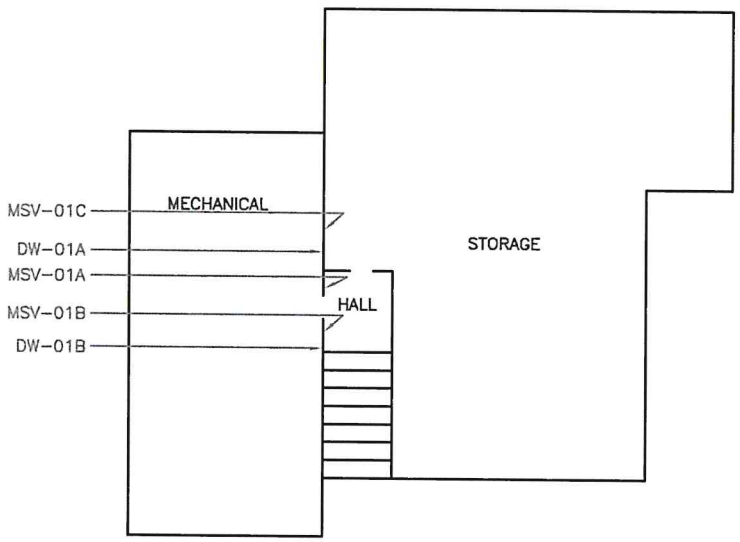


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
VISITOR INFORMATION CENTER – 2ND FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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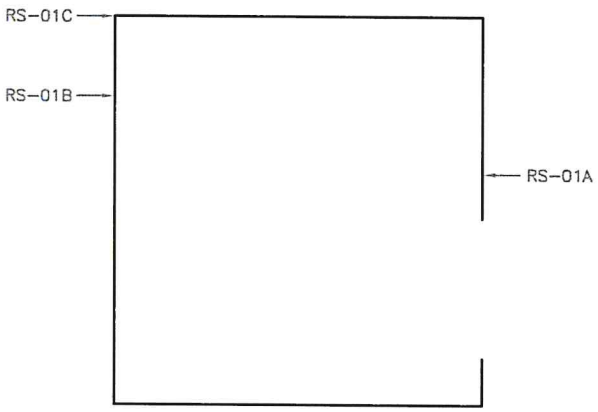


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
VISITOR INFORMATION CENTER – WELL/PUMP HOUSE
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
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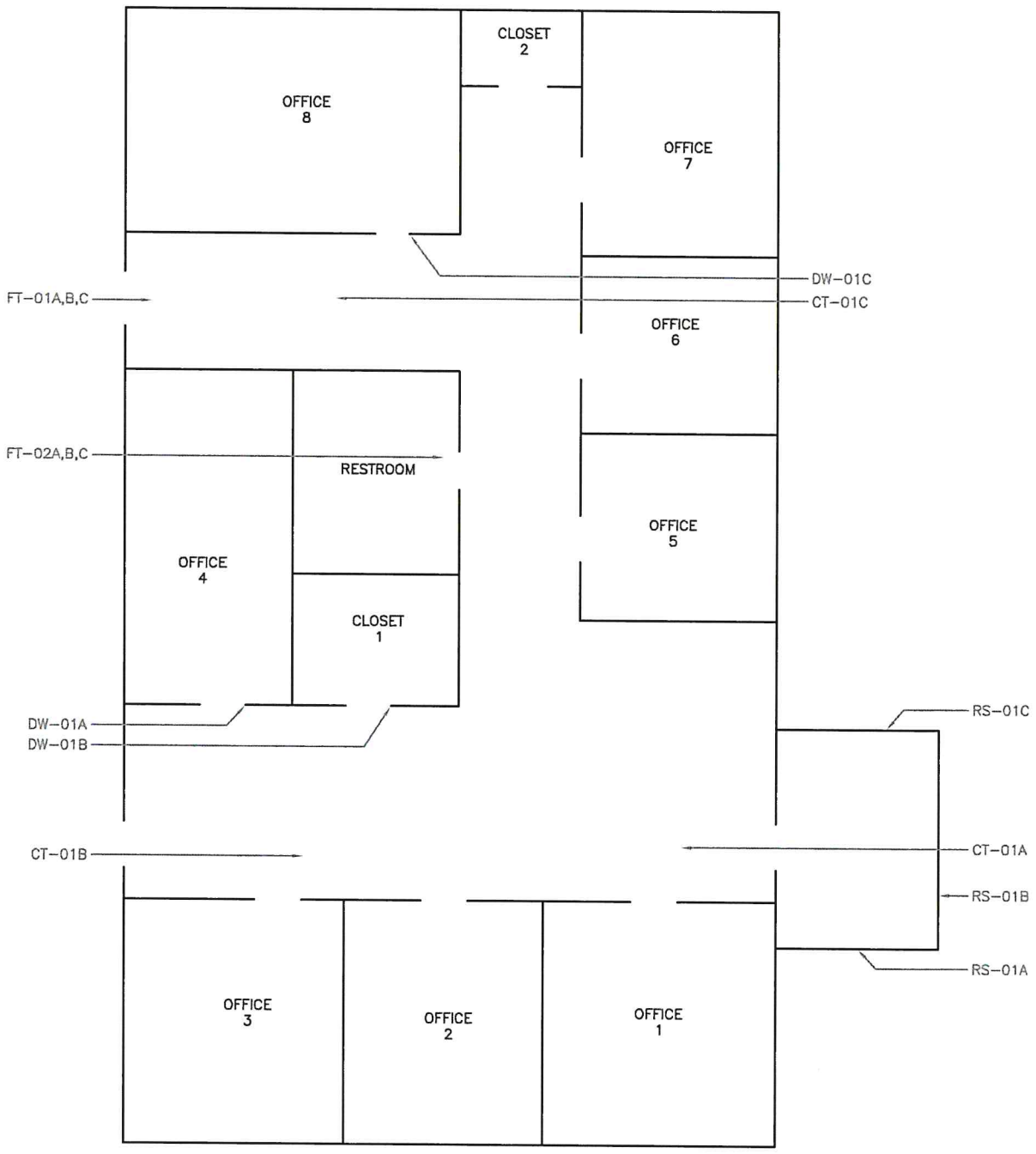


ASBESTOS CONTAINING MATERIALS LEGEND

RUBBER MEMBRANE ROOF ASSUMED TO CONTAIN ASBESTOS.

GENERAL LEGEND

FT–11B: POSITIVE SAMPLE LOCATION

CT–01A: NEGATIVE SAMPLE LOCATION

N

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# ASBESTOS SAMPLE LOCATION PLAN
OFFICE TRAILER
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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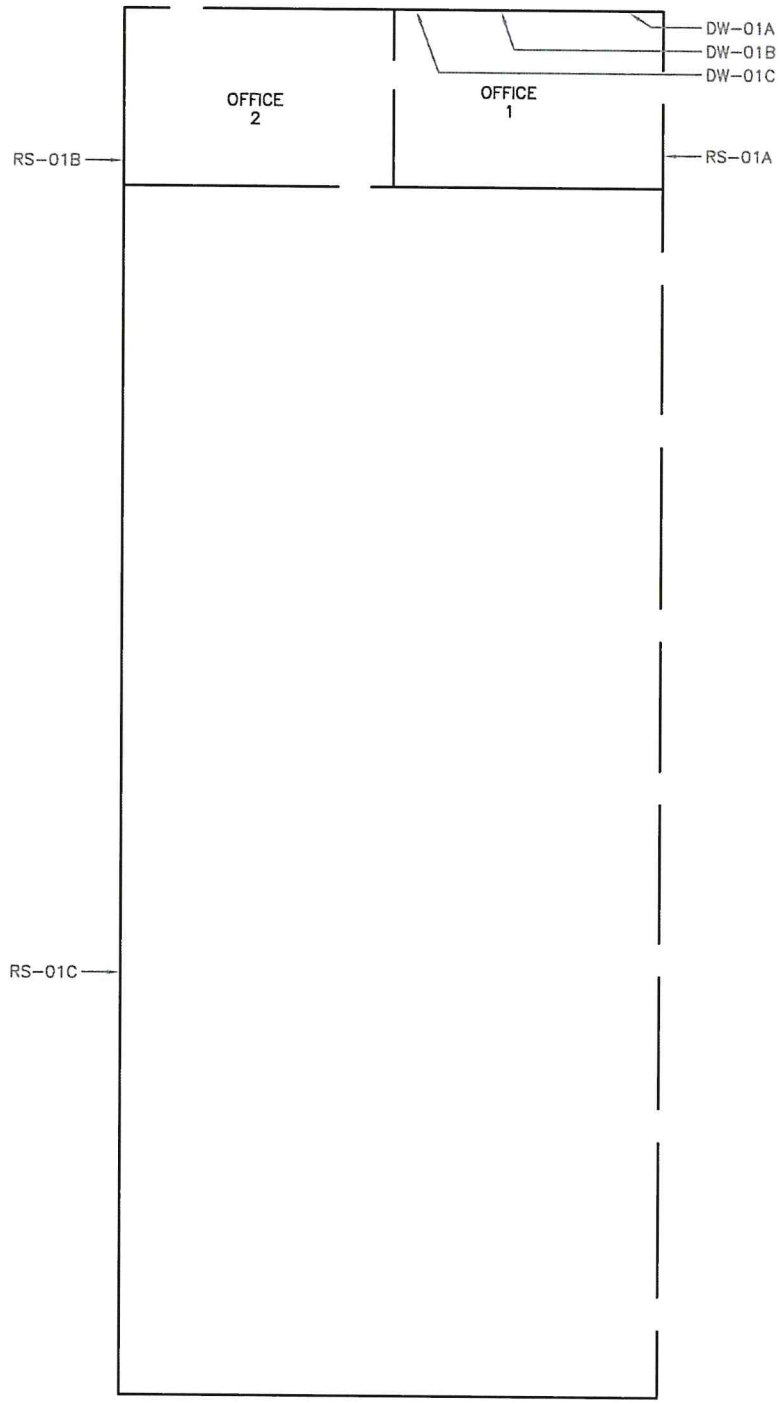


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
LIGHT ENGINE EQUIPMENT STORAGE
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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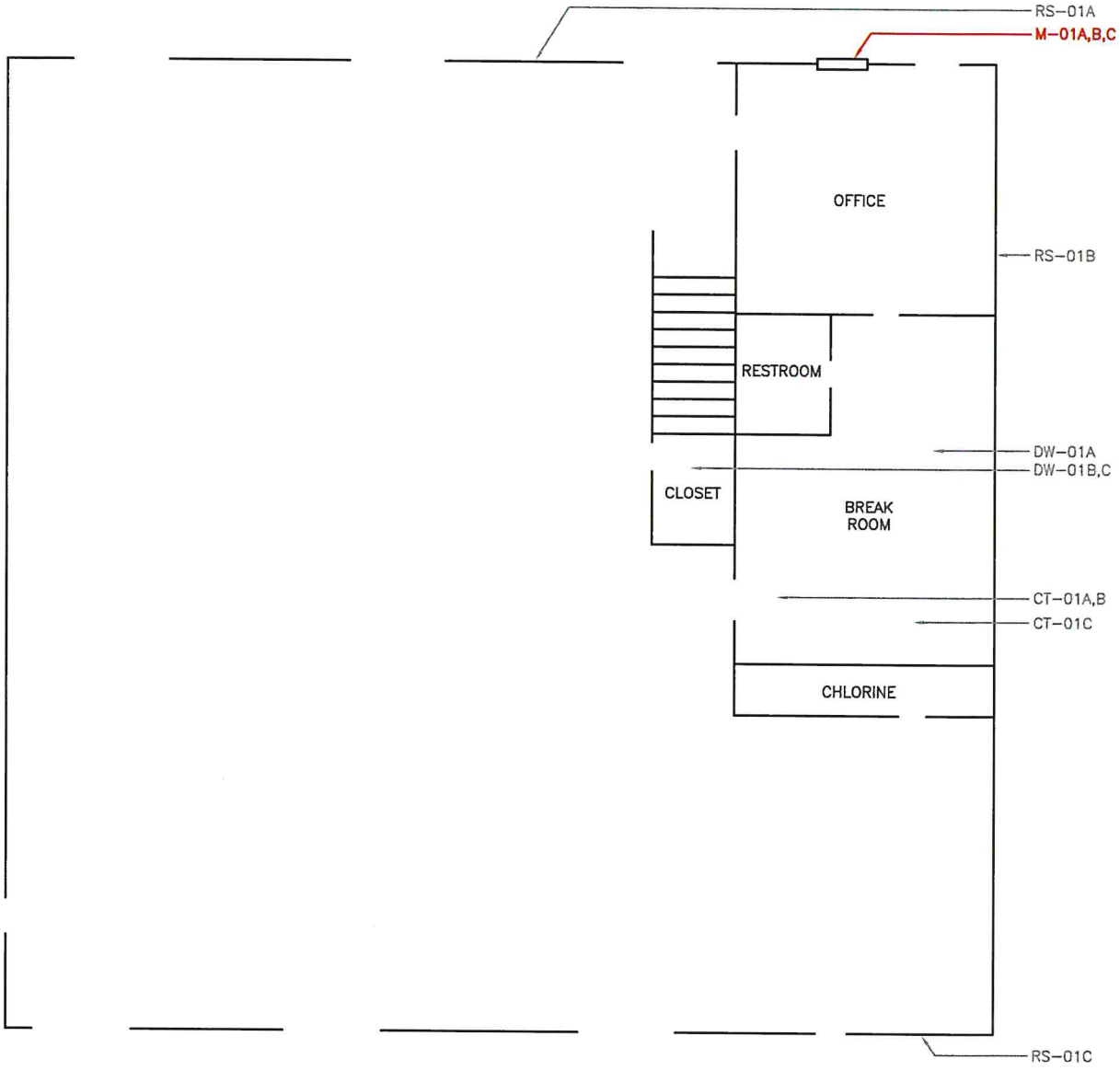


ASBESTOS CONTAINING MATERIALS LEGEND

M-01: WINDOW GLAZING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
MAINTENANCE SHOP – 1ST FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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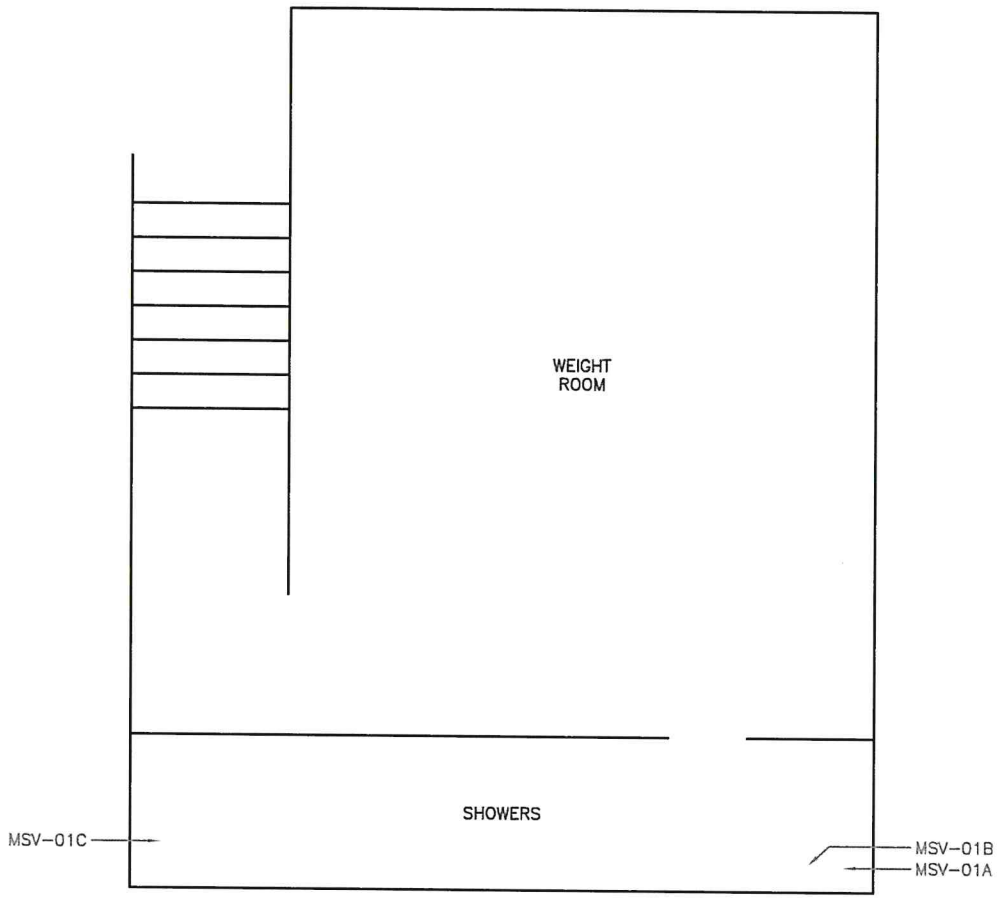


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
MAINTENANCE SHOP – 2ND FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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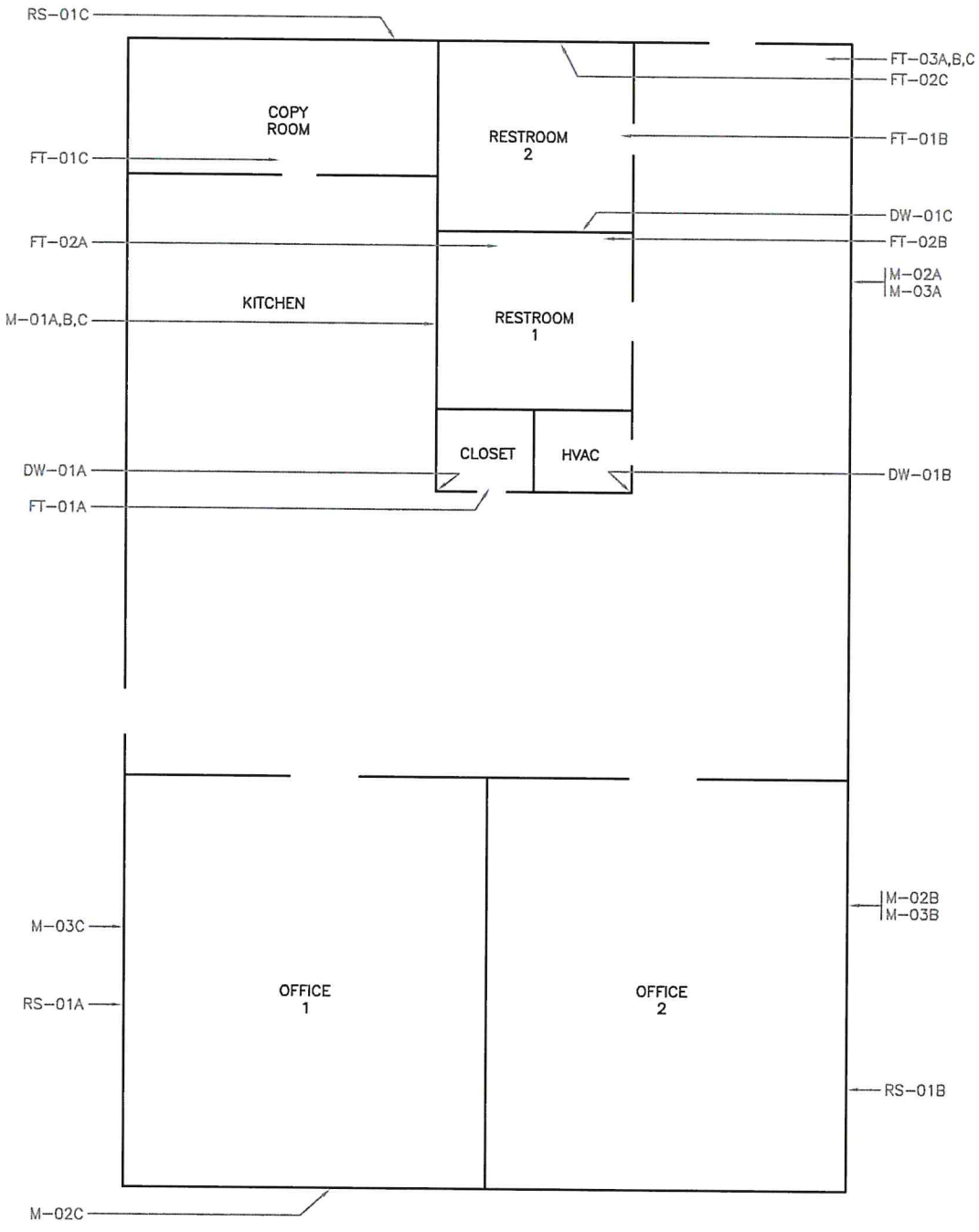


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
FIRE CACHE/ADMINISTRATION BLDG.
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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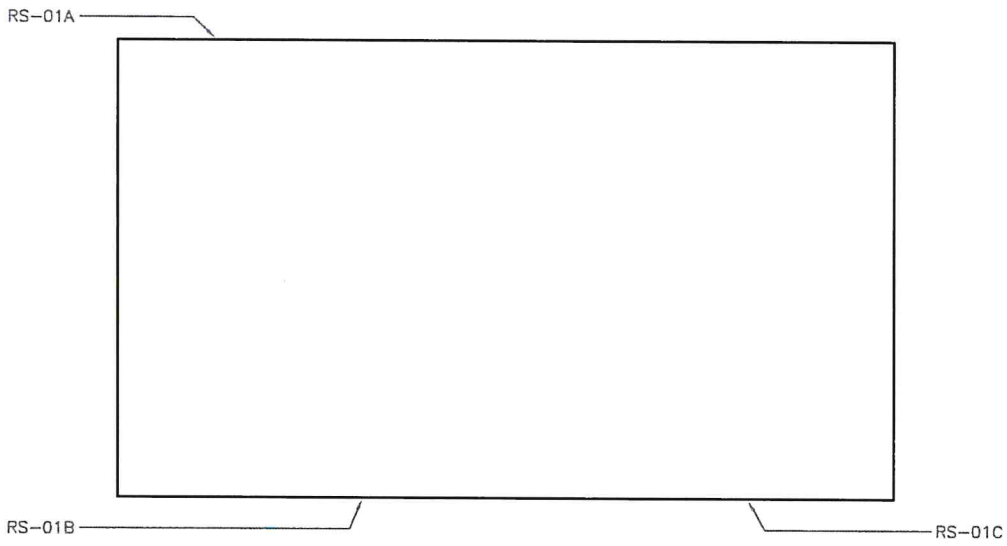


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT–11B: POSITIVE SAMPLE LOCATION

CT–01A: NEGATIVE SAMPLE LOCATION

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
FRONT POLE SHED
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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TITUSVILLE, FL

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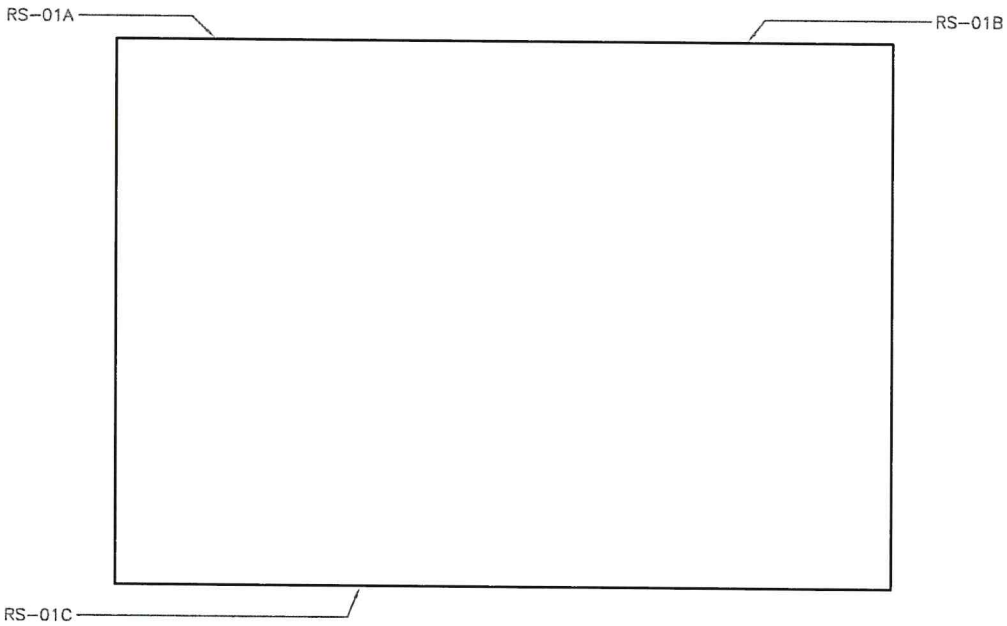


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
BACK POLE SHED
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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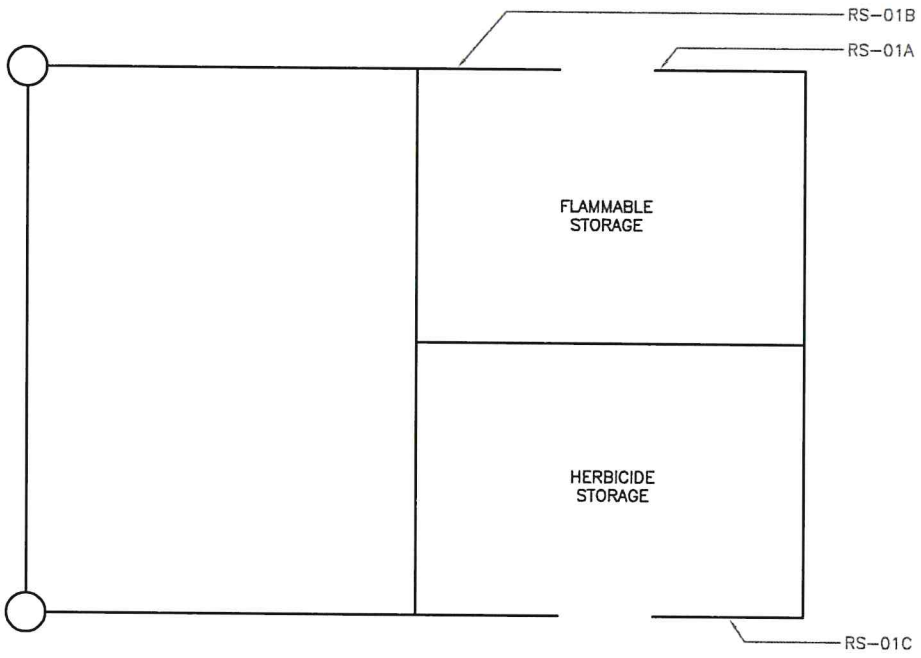


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
FUEL ISLAND BUILDING
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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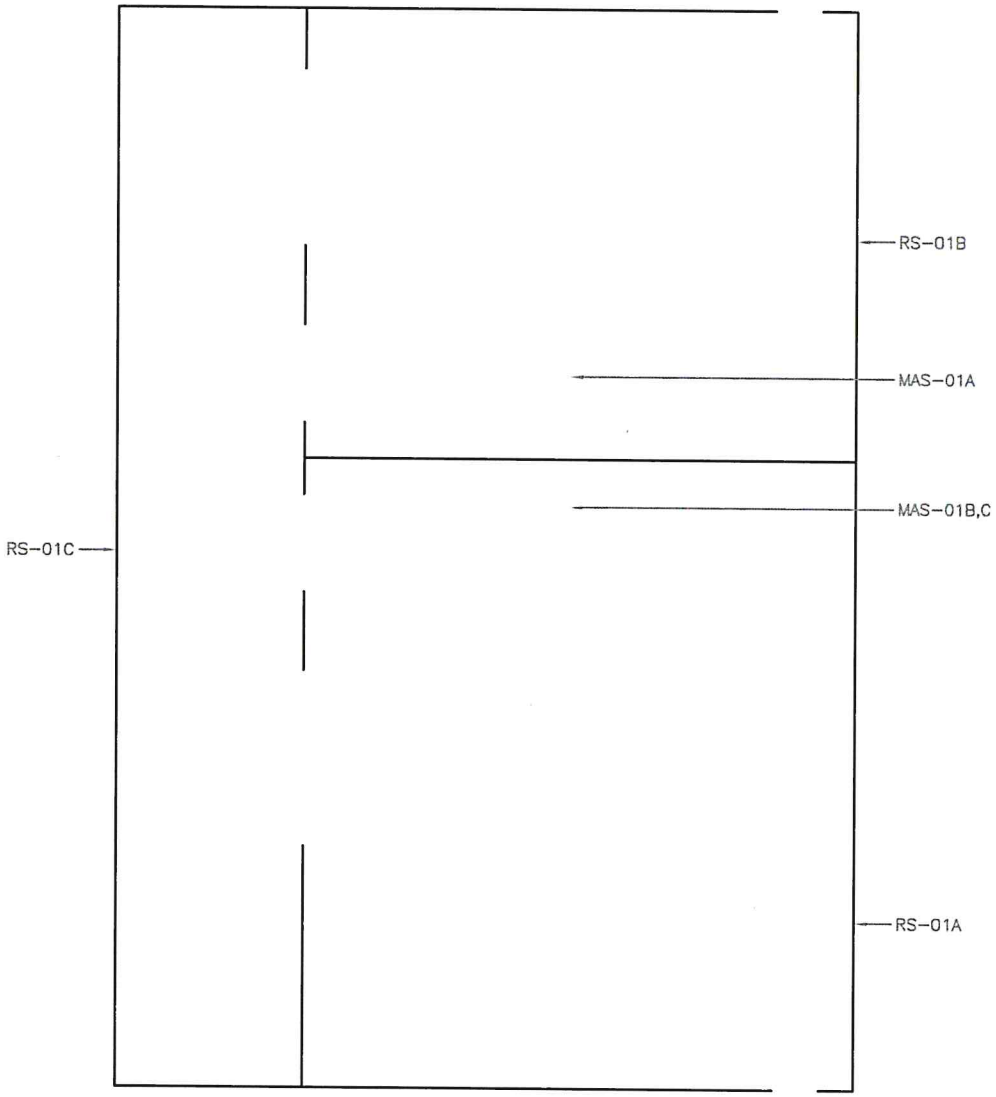


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

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# ASBESTOS SAMPLE LOCATION PLAN
WAREHOUSE
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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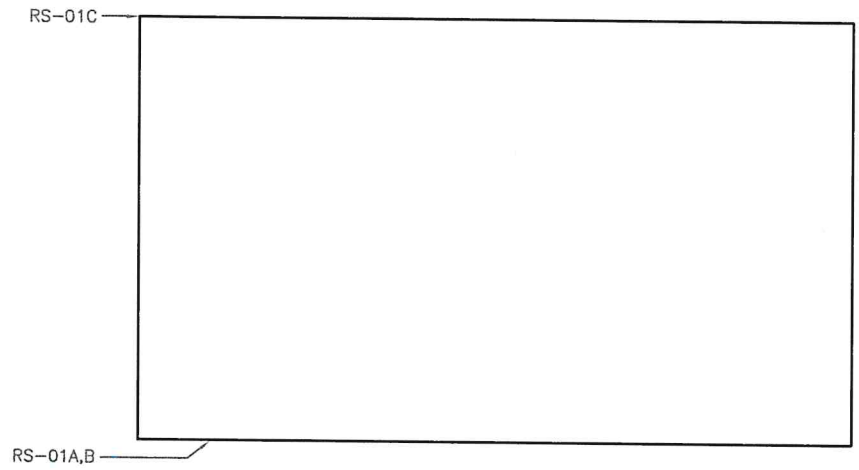


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT-11B: POSITIVE SAMPLE LOCATION

CT-01A: NEGATIVE SAMPLE LOCATION

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
ENGINE FILL STATION
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

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TITUSVILLE, FL

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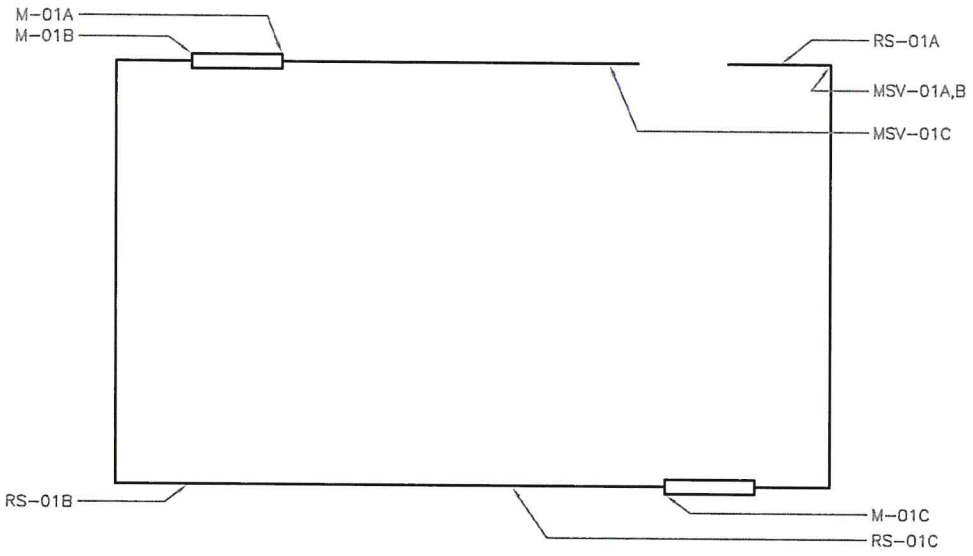


ASBESTOS CONTAINING MATERIALS LEGEND

NONE OF THE SAMPLES COLLECTED WERE DETERMINED TO BE ASBESTOS CONTAINING.

GENERAL LEGEND

FT–11B: POSITIVE SAMPLE LOCATION

CT–01A: NEGATIVE SAMPLE LOCATION

N

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ASBESTOS SAMPLE LOCATION PLAN
PILOT BRIEFING/OFFICE BLDG.
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

Prepared for:
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TITUSVILLE, FL

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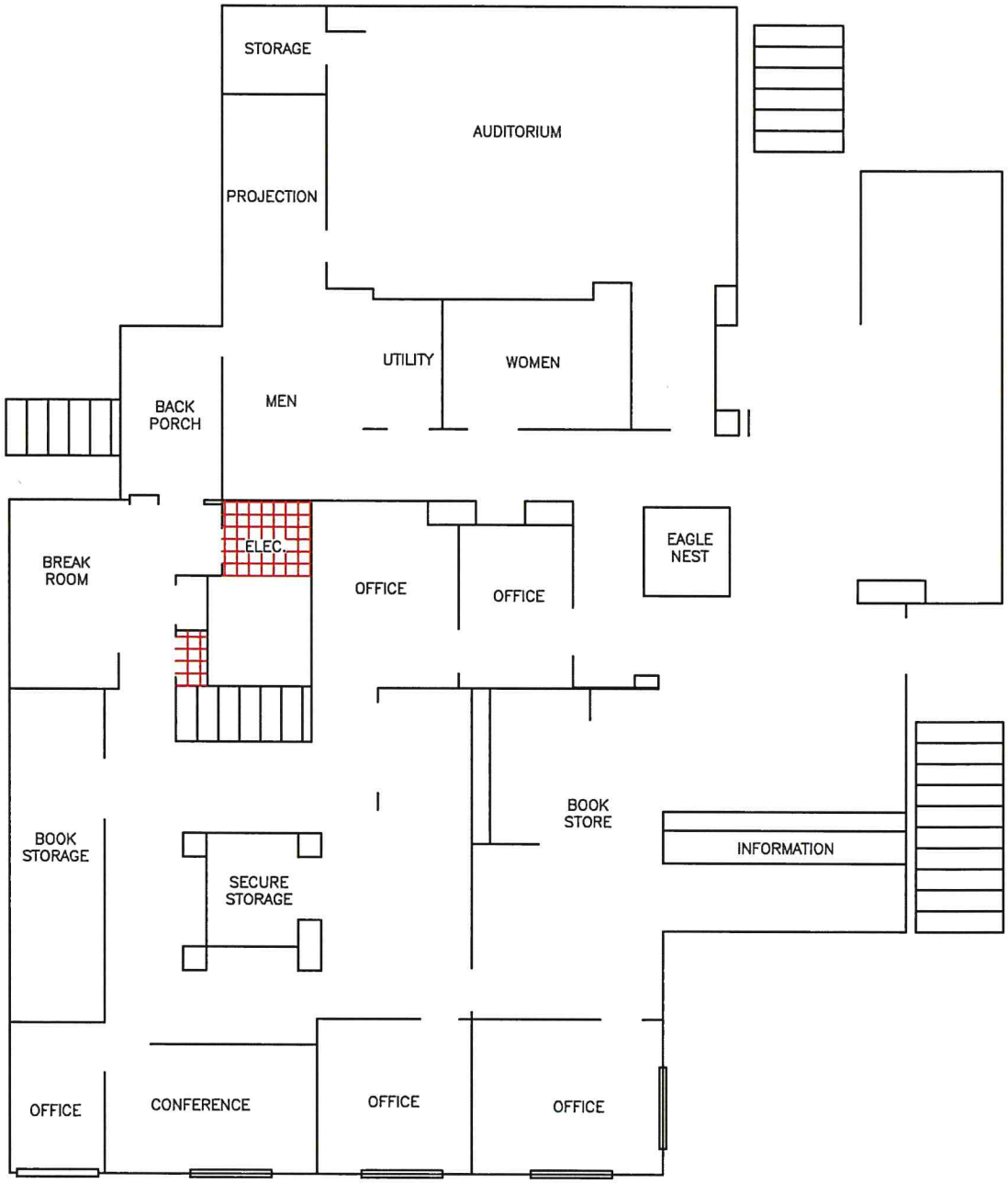
DATE: 12/01/11
OF 15 SHEET(S)

# APPENDIX D
# Asbestos Location Plans




ASBESTOS CONTAINING MATERIALS LEGEND

BROWN PEBBLE SHEET VINYL FLOORING.

N

THIS DRAWING IS NOT TO SCALE

# ASBESTOS LOCATION PLAN

VISITOR INFORMATION CENTER – 1ST FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

Prepared for:
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| GLE CAD NO. | CAD/PROJ/11463/00002 |
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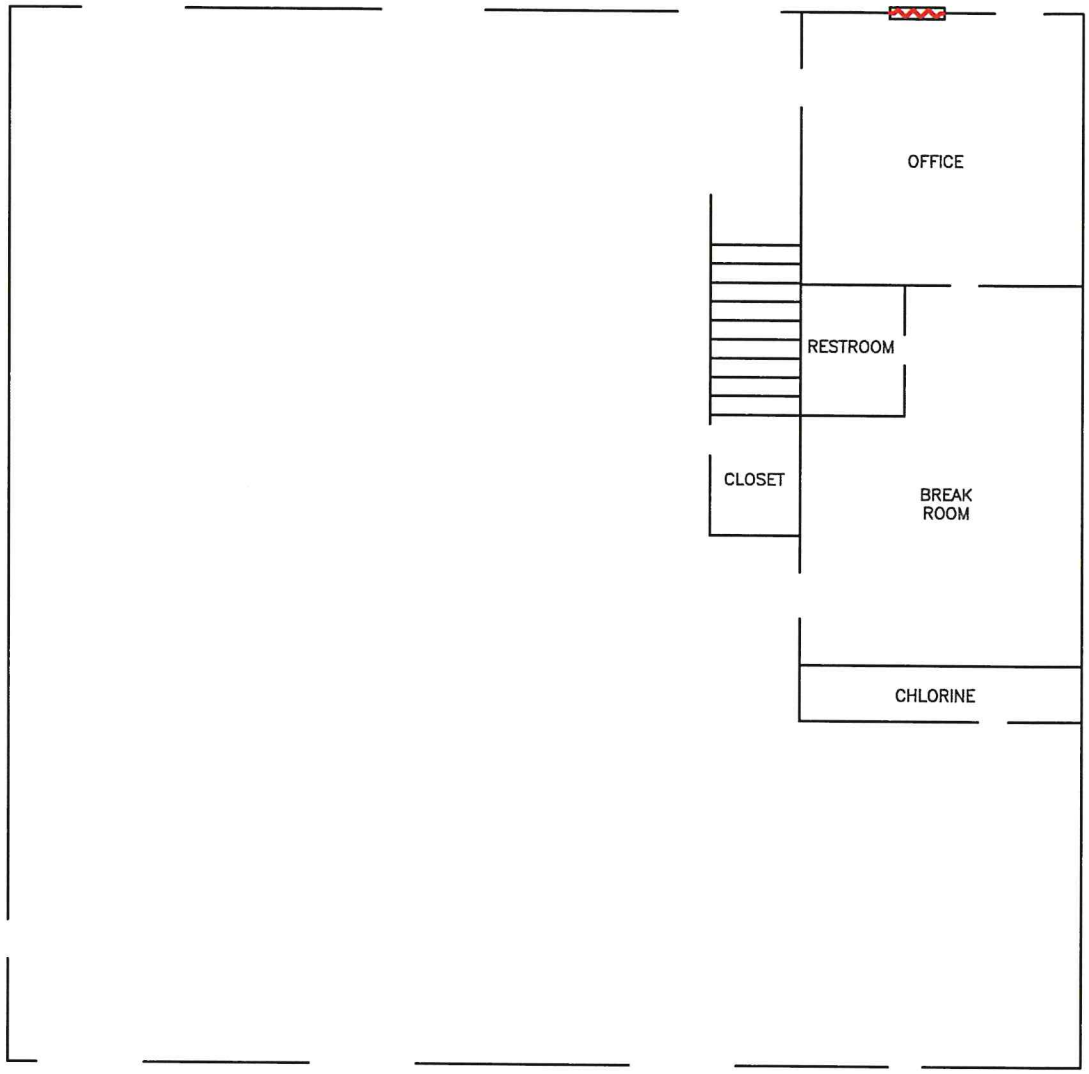
| CHECKED: TT | SHEET: AL-1 |
| DATE: 12/01/11 | OF 2 SHEET(S) |




ASBESTOS CONTAINING MATERIALS LEGEND

WINDOW GLAZING.

N

THIS DRAWING IS NOT TO SCALE

ASBESTOS SAMPLE LOCATION PLAN
MAINTENANCE SHOP – 1ST FLOOR
MERRITT ISLAND NATIONAL WILDLIFE REFUGE
SR 402 (5 MI EAST OF TITUSVILLE, FL)
MERRIT ISLAND, FL

Prepared for:
U.S. FISH & WILDLIFE SERVICES
P.O. BOX 2683
TITUSVILLE, FL

Prepared By: GLE ASSOCIATES, INC.
4300 W CYPRESS STREET STE 400
TAMPA, FLORIDA 33607
(813) 241-8350 (813) 241-8737

GLE
Facilities and Environmental
Consultants

| | |
|---|---|
| GLE CAD NO. | CAD/PROJ/11463/00002 |
| DRAWN: BPH | JOB NO.: 11463-00002 |
| CHECKED: TT | SHEET: AL-2 |
| DATE: 12/01/11 | OF 2 SHEET(S) |

# APPENDIX E
# Photographs

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Biolab Well-Pump House

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

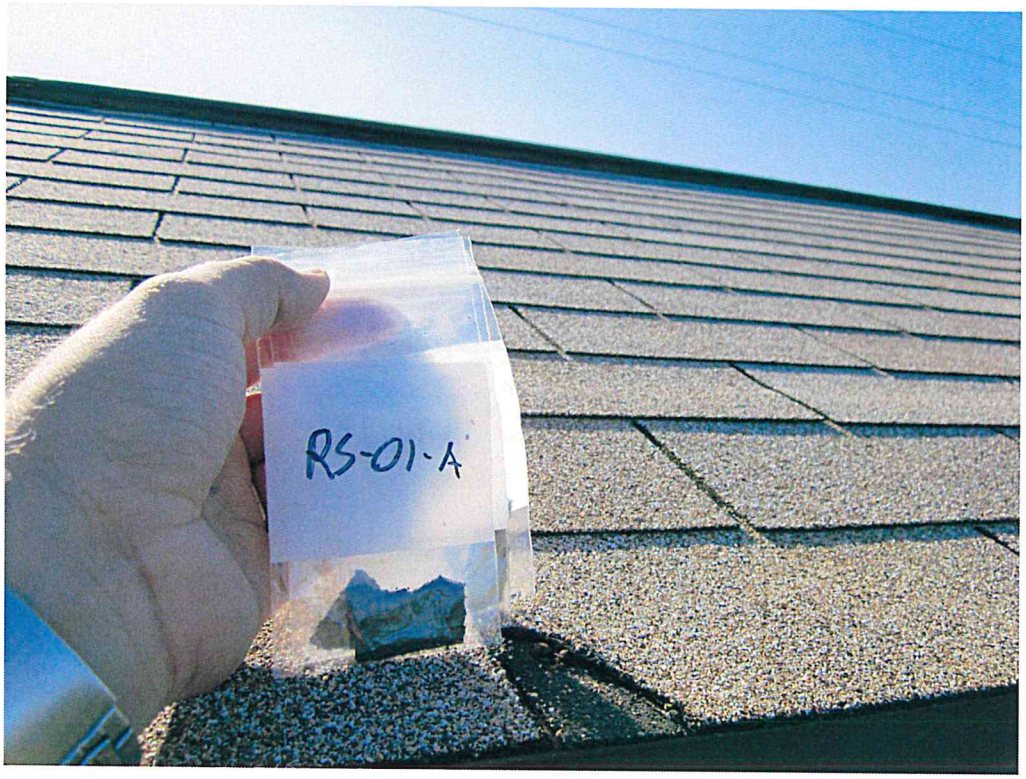



**Figure P-2**
Upper Photo: Roof Shingles

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Biolab Well-Pump House

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rent Elevation

November 21, 2011

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8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Visitor Information Center

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**

Upper Photo: Drywall and joint compound.
Lower Photo: Beige Square Sheet Vinyl Flooring.

November 21, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Visitor Information Center

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

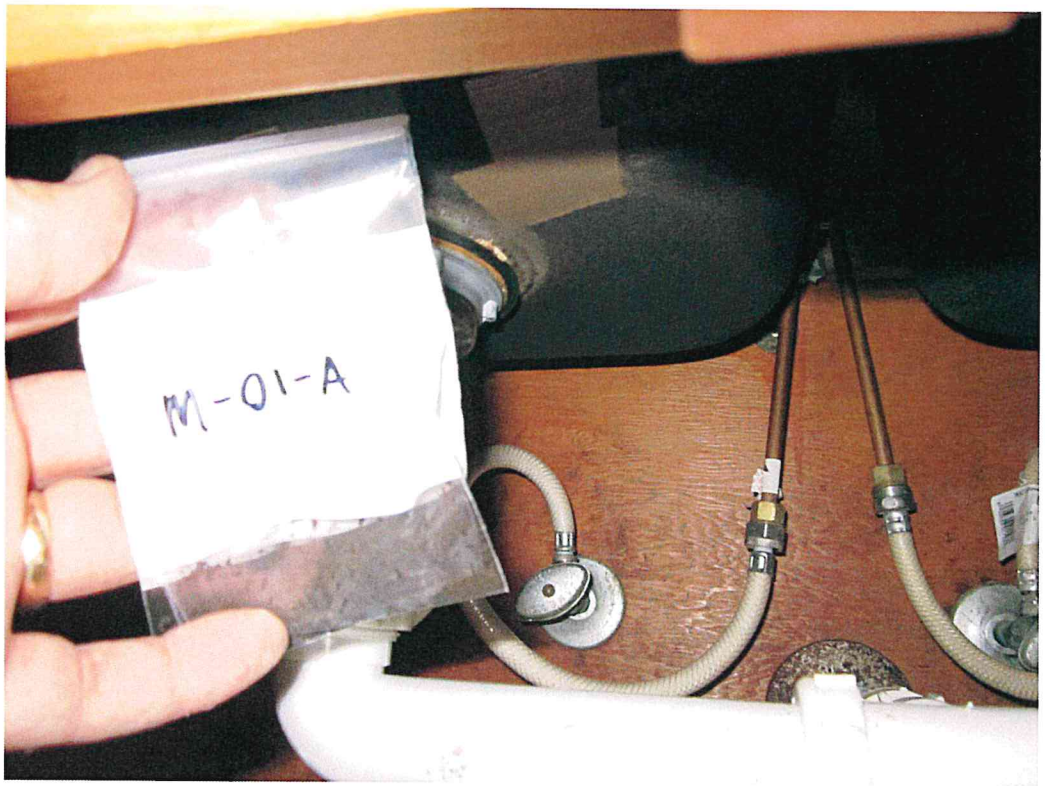


**Figure P-3**

Upper Photo: Brown pebble Sheet Vinyl Flooring.
Lower Photo: Black sink undercoat.

November 21, 2011

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Jacksonville, FL 32256 (904) 296-1880

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Visitor Information Center

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-3 |
| Date | |
| 11/30/11 | |

**Figure P-4**
Upper Photo: Window Caulk.
Lower Photo: Roofing Shingles.

November 21, 2011

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8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

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Visitor Information Center

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-4 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 21, 2011

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8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

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Visitor Center Well - Pump House

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

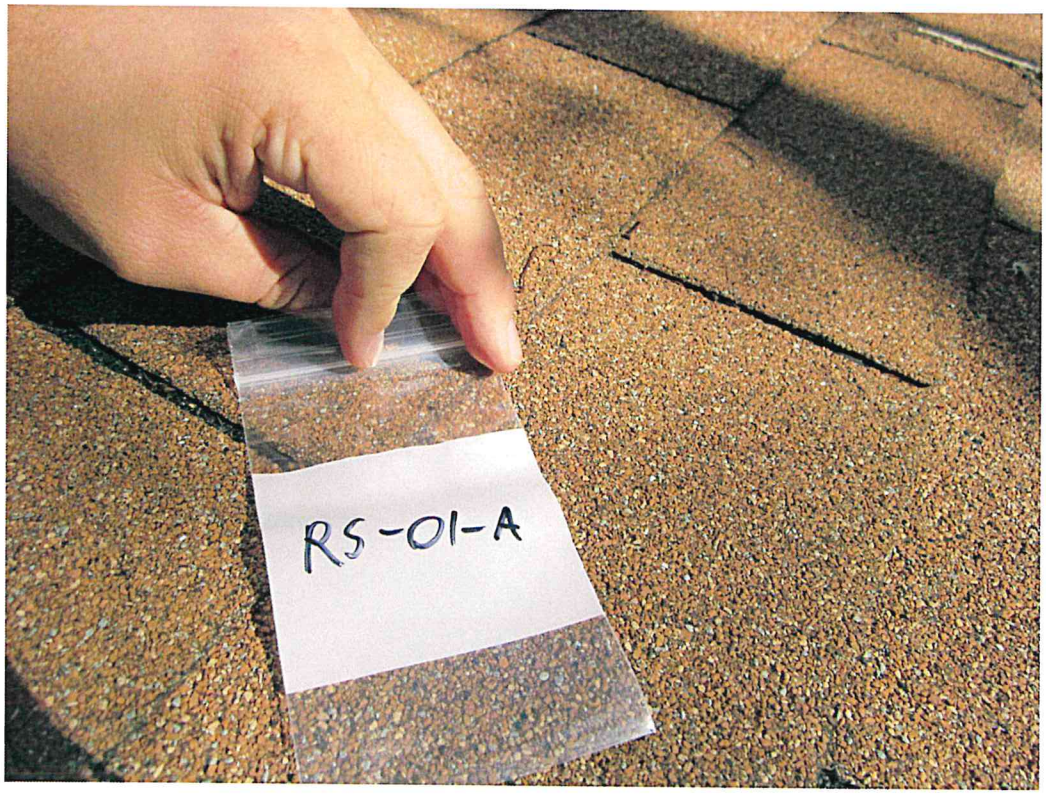


**Figure P-2**

Upper Photo: Roof Shingles.

November 21, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

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Visitor Center Well - Pump House

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

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Jacksonville, FL 32256 (904) 296-1880

GLE

| Office Trailer | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-1 |
| Date: 11/30/11 | |

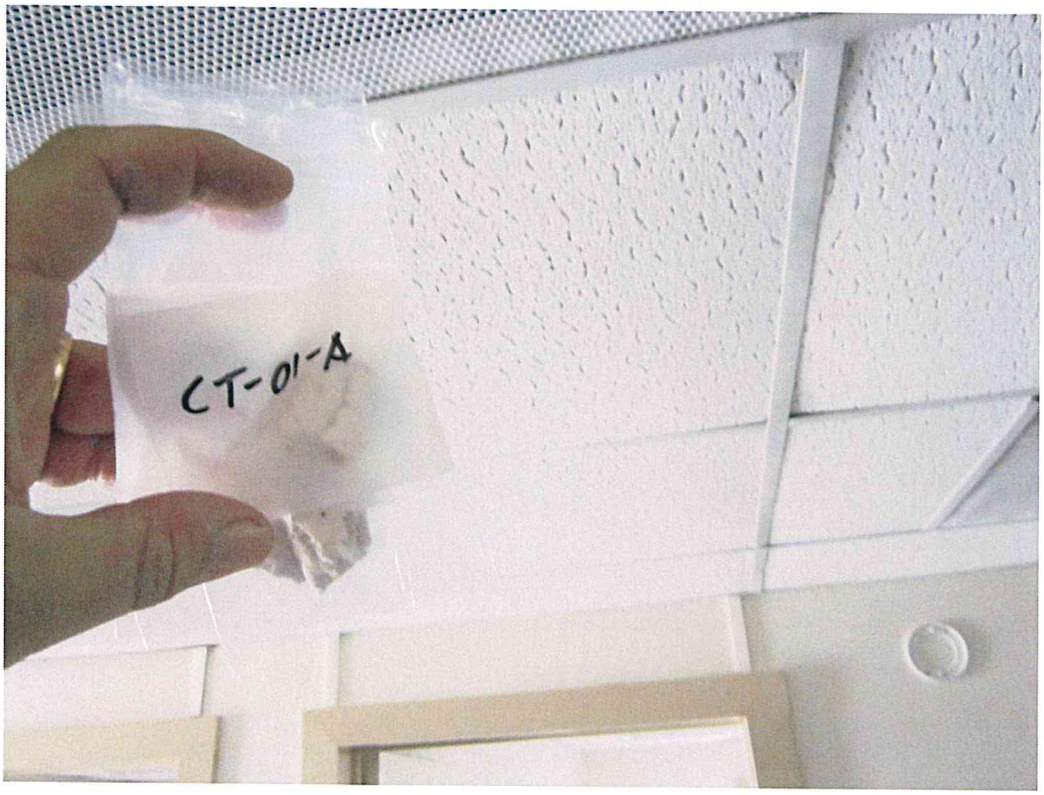


**Figure P-2**

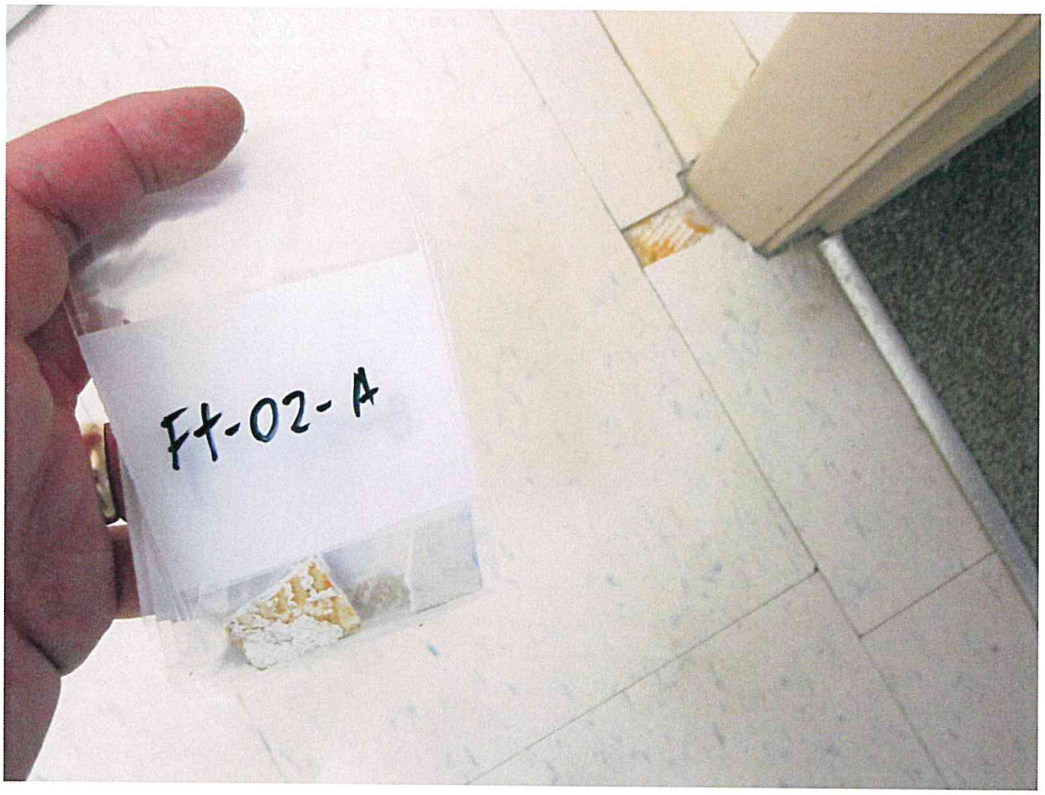
Upper Photo: 2x4 worm hole patterned ceiling tile.
Lower Photo: 12x12 tan floor tile.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Office Trailer | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-2 |
| Date: 11/30/11 | |



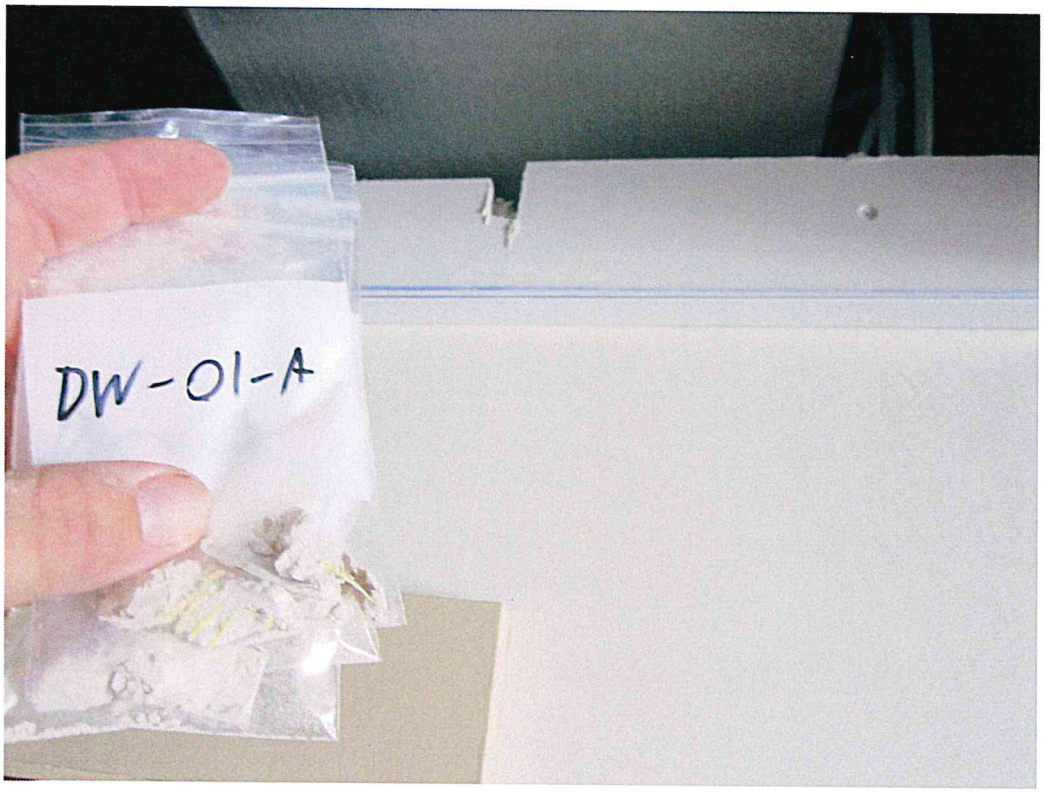




**Figure P-3**

Upper Photo: 12x12 cream & blue flake floor tile.
Lower Photo: Drywall & joint compound.

November 22, 2011

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Jacksonville, FL 32256 (904) 296-1880

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Office Trailer

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-3 |
| Date | |
| 11/30/11 | |

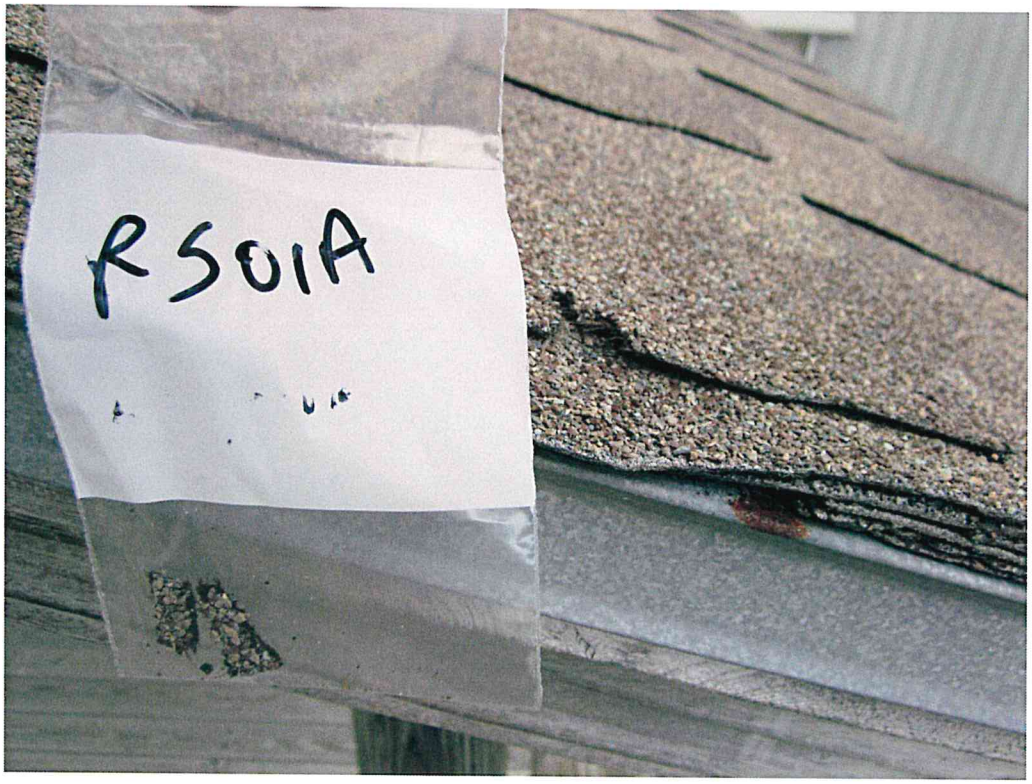


**Figure P-4**

Upper Photo: Roof shingles.

November 22, 2011

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Jacksonville, FL 32256 (904) 296-1880

GLE

Office Trailer

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-4 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Light Engine Equipment Storage

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

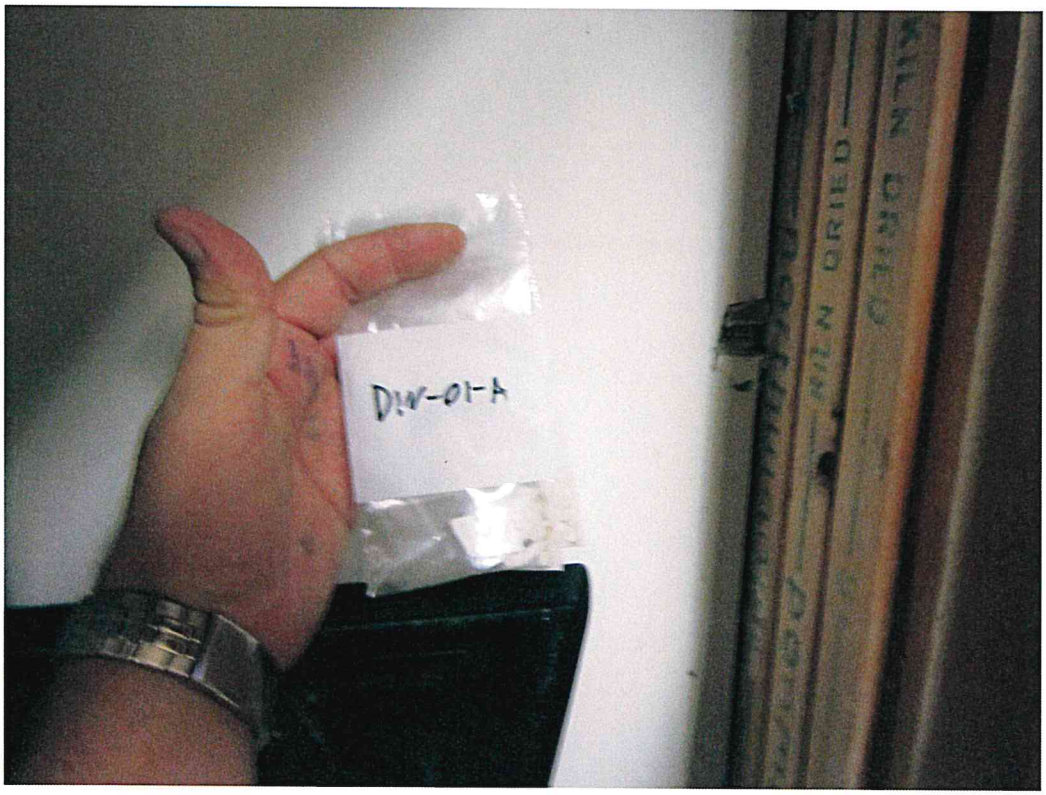



**Figure P-2**

Upper Photo: Roof shingles.
Lower Photo: Drywall and joint compound.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Light Engine Equipment Storage

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

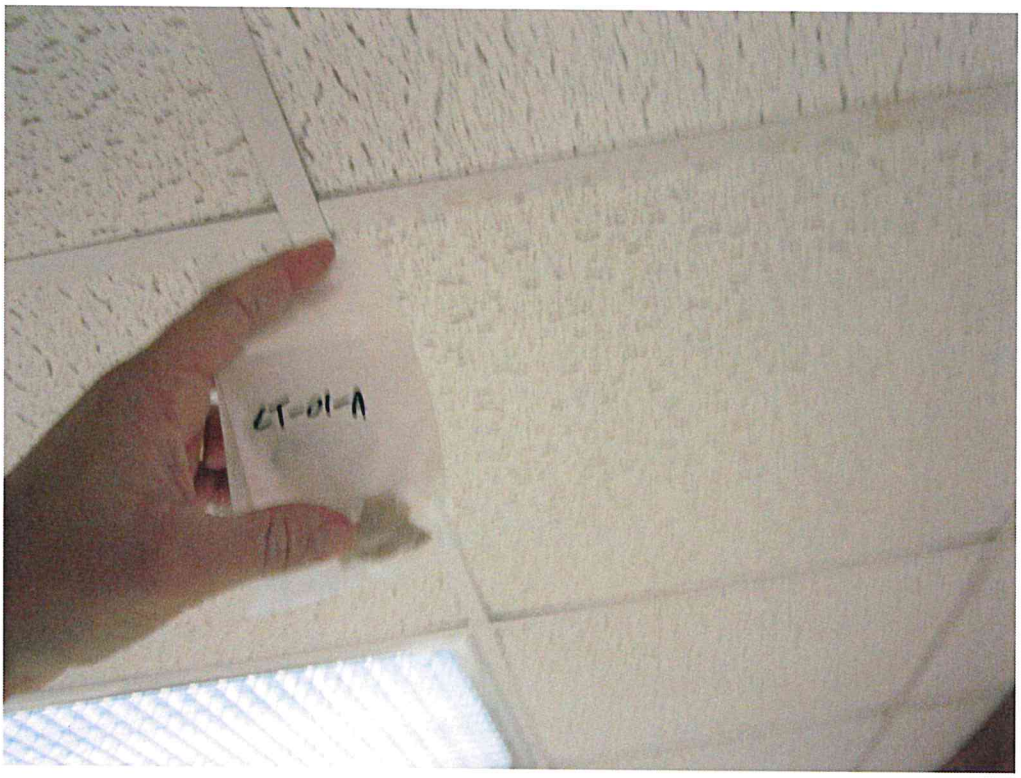
**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Maintenance Shop | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-1 |
| Date: 11/30/11 | |



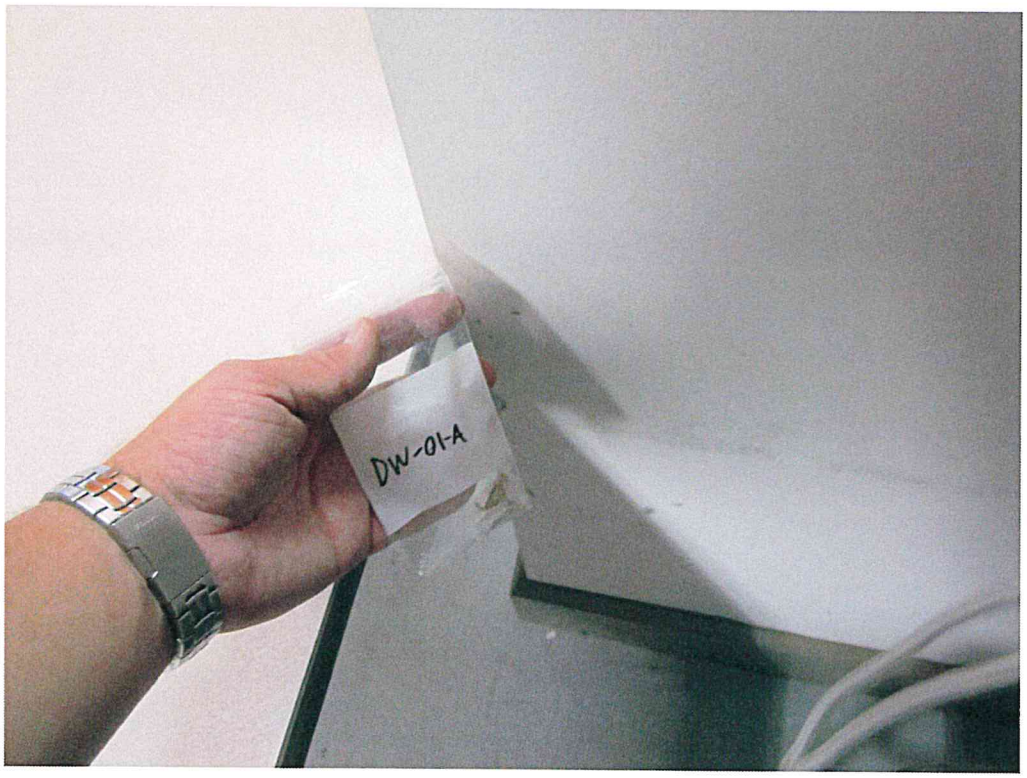




**Figure P-2**

Upper Photo: 2x2 worm hole patterned ceiling tile.
Lower Photo: Drywall and joint compound.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Maintenance Shop | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-2 |
| Date: 11/30/11 | |

**Figure P-3**

Upper Photo: Tan & blue patterned sheet vinyl flooring.
Lower Photo: Roofing shingles.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

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Maintenance Shop

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-3 |
| Date | |
| 11/30/11 | |

**Figure P-4**
Upper Photo: Window Glazing.

November 22, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

GLE

Maintenance Shop

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-4 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Fire Cache / Admin. Bldg

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**
Upper Photo: White sink undercoat.
Lower Photo: White 12x12 floor tile

November 22, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

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| Fire Cache / Admin. Bldg | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-2 |
| Date: 11/30/11 | |

**Figure P-3**

Upper Photo: Drywall & joint compound.
Lower Photo: 12x12 tan with brown flake floor tile.

November 22, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

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Fire Cache / Admin. Bldg

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-3 |
| Date | |
| 11/30/11 | |

**Figure P-4**
Upper Photo: 12x12 brown small square floor tile.
Lower Photo: Exterior window caulk.

November 22, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

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Fire Cache / Admin. Bldg

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-4 |
| Date | |
| 11/30/11 | |

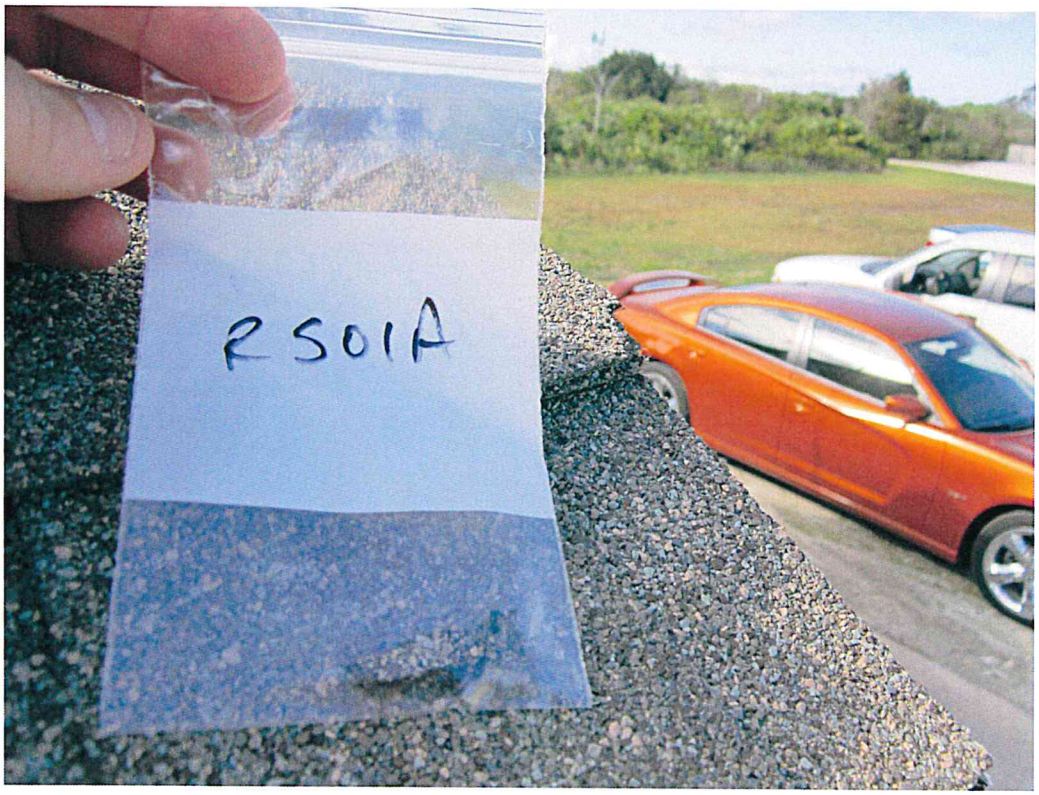


**Figure P-5**
Upper Photo: Caulk on hurricane shutters.
Lower Photo: Roofing shingles.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Fire Cache / Admin. Bldg | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-5 |
| Date: 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 21, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

GLE

| Front Pole Shed | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-1 |
| Date: 11/30/11 | |

**Figure P-2**
Upper Photo: Roof shingles

November 21, 2011

Prepared By: GLE Associates, Inc.
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Jacksonville, FL 32256 (904) 296-1880

GLE

| Front Pole Shed | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-2 |
| Date: 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 21, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Back Pole Shed

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**
Upper Photo: Roof shingles

November 21, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Back Pole Shed

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Fuel Island Building

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**
Upper Photo: Roof shingles

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Fuel Island Building

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-1**

Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Warehouse | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-1 |
| Date: 11/30/11 | |

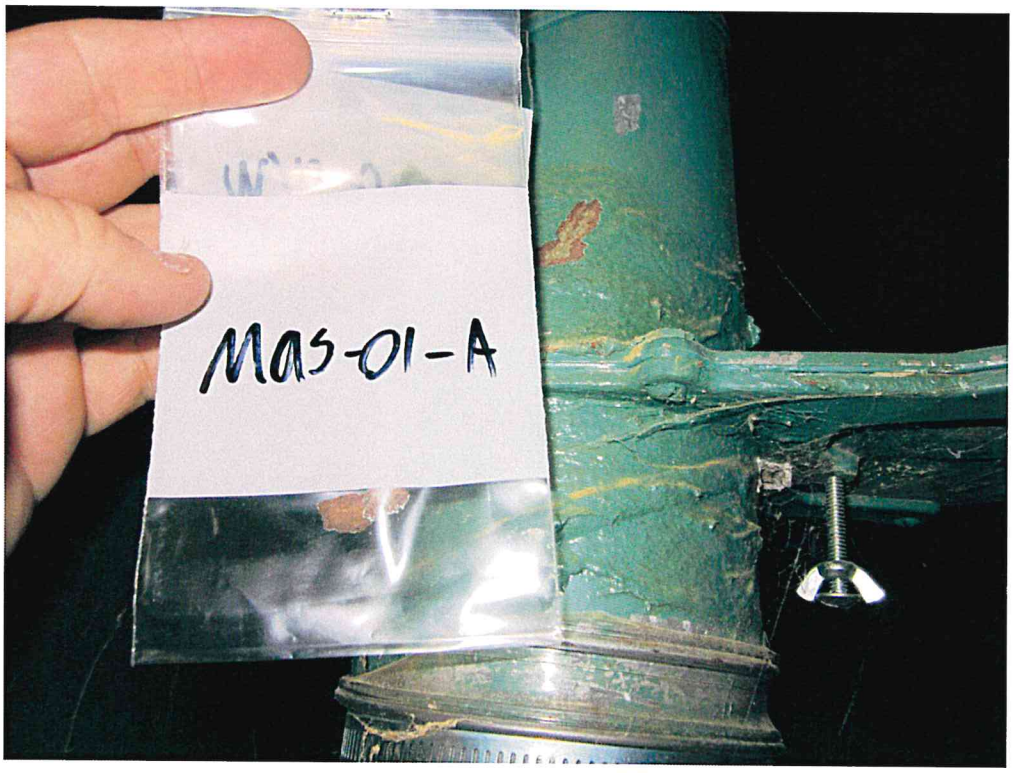

**Figure P-2**
Upper Photo: Pipe Mastic.
Lower Photo: Roofing Shingles.

November 22, 2011

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Jacksonville, FL 32256 (904) 296-1880

GLE

| Warehouse | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-2 |
| Date: 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 21, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Engine Fill Station

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**
Upper Photo: Roof shingles

November 21, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Engine Fill Station

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Pilot Briefing Building

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |

**Figure P-2**

Upper Photo: Roofing shingles.
Lower Photo: Blue speck patterned sheet vinyl flooring.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Pilot Briefing Building

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-2 |
| Date | |
| 11/30/11 | |

**Figure P-3**
Upper Photo: Window caulking.

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

| Pilot Briefing Room | |
|---|---|
| Drawn: THT | Job No.: 11463-00002 |
| Checked: MBC | Figure: P-3 |
| Date: 11/30/11 | |

**Figure P-1**
Upper Photo: Front Elevation
Lower Photo: Rear Elevation

November 22, 2011

Prepared By: GLE Associates, Inc.
8659 Baypine Road, Suite 306
Jacksonville, FL 32256 (904) 296-1880

GLE

Storage Shed at Tranquility

| Drawn | Job No. |
|---|---|
| THT | 11463-00002 |
| Checked | Figure |
| MBC | P-1 |
| Date | |
| 11/30/11 | |